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(54) **MALWARE MONITORING AND DETECTION**

(71) Applicant: **Rubrik, Inc.**, Palo Alto, CA (US)

(72) Inventors: **Gregory Robert Johnston**, Mountain View, CA (US); **Ishaan Sang**, Mountain View, CA (US); **Kunal Sean Munshani**, Fremont, CA (US); **Adam Gee**, Palo Alto, CA (US); **Guilherme Menezes**, San Jose, CA (US); **Kelvin Kwan**, San Mateo, CA (US); **Shivanshu Agrawal**, Sunnyvale, CA (US); **Muraliraja Muniraju**, Fremont, CA (US); **Aaron Chen**, Cupertino, CA (US)

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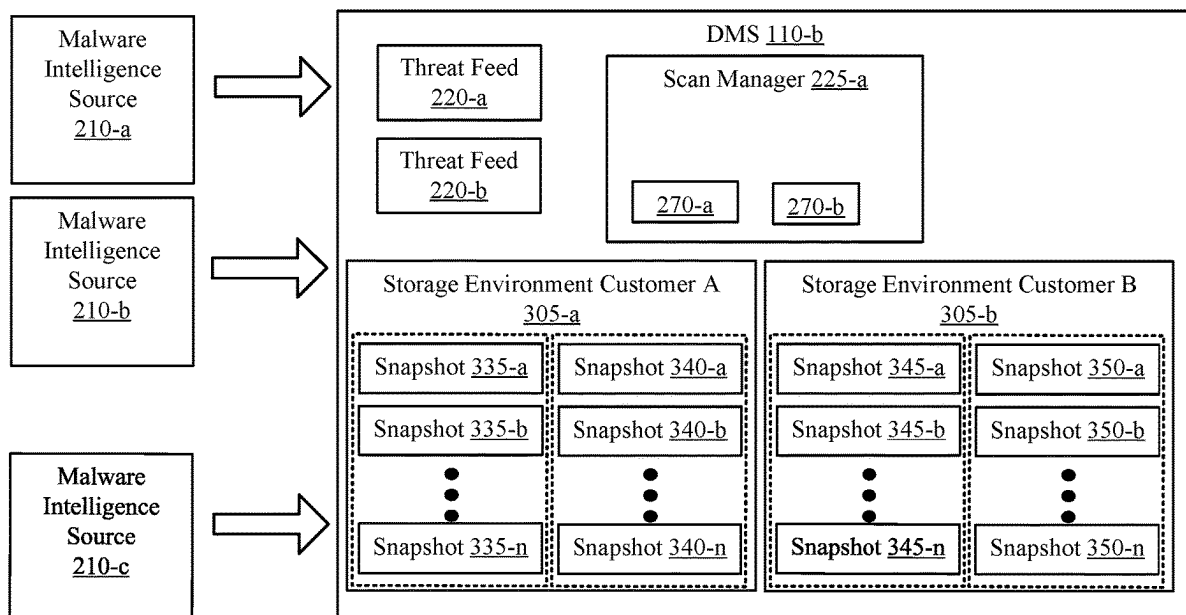
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(57)

ABSTRACT

Methods, systems, and devices for data management are described. A data management system (DMS) may maintain a malware threat feed that includes updated (e.g., periodically or continuously updated) malware intelligence. For example, the malware intelligence may involve YARA rules and known hashes indicative of malware. The DMS may use the malware intelligence to scan each new snapshot of a customer computing system to determine if the new snapshot contains malware. The DMS may periodically, or in response to reception of new malware intelligence, check the files within the most recent snapshot of each computing system for a customer to determine whether any computing system of the customer contains malware. When malware is detected in a file within a snapshot—either based on checking a newly captured snapshot or when checking snapshots based on new malware intelligence—the DMS may determine the temporally first snapshot which contained the corrupted file.



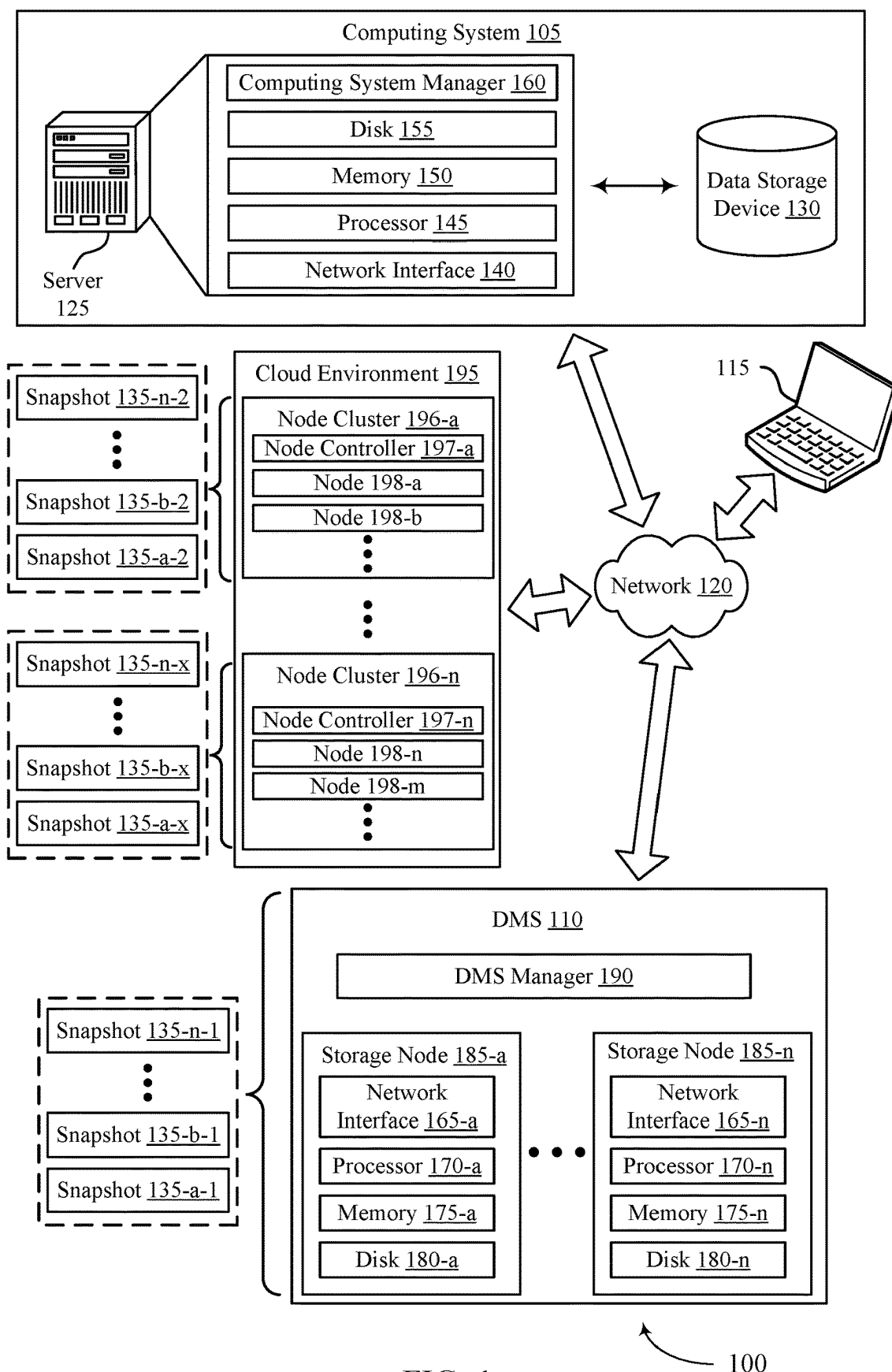


FIG. 1

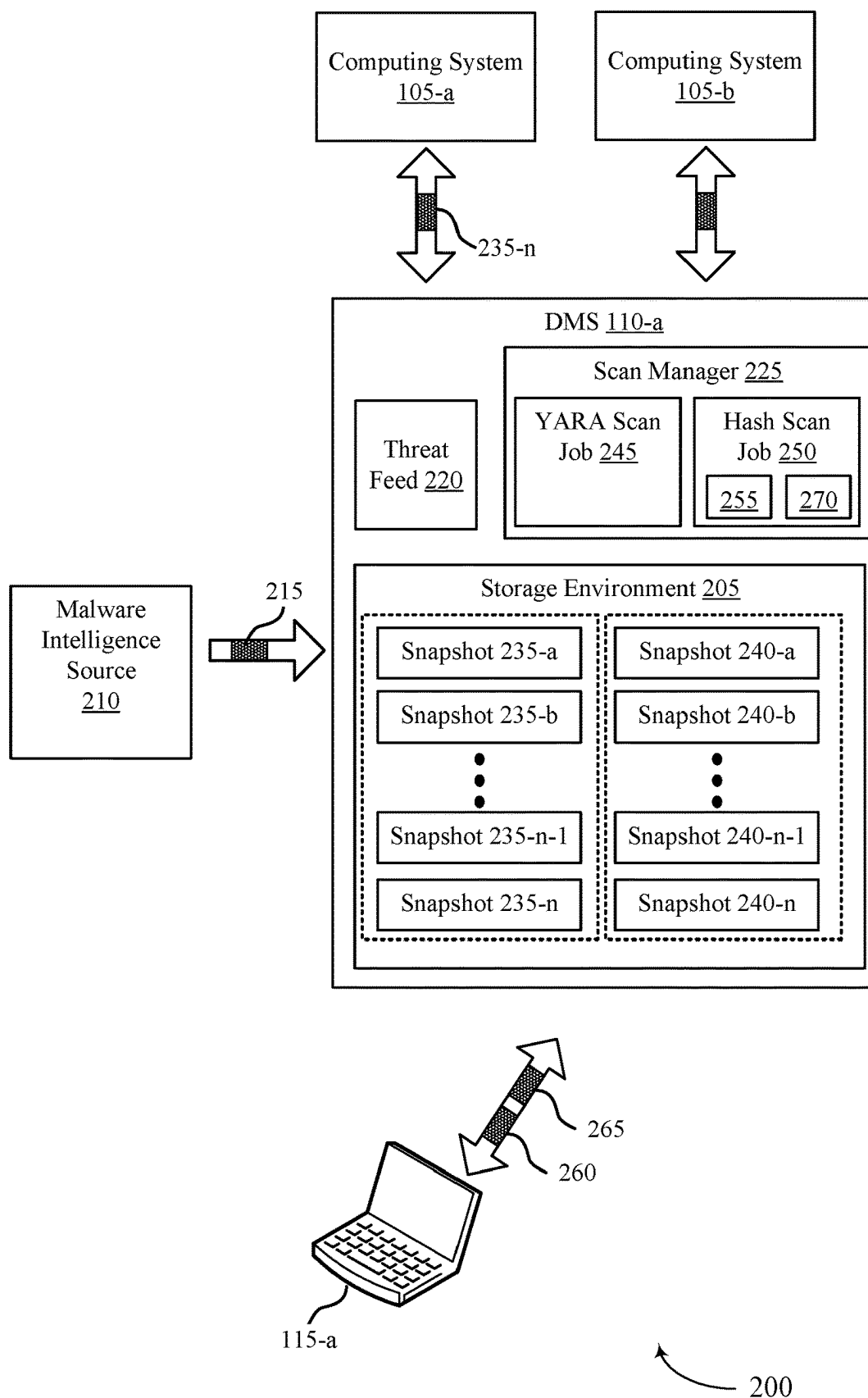


FIG. 2

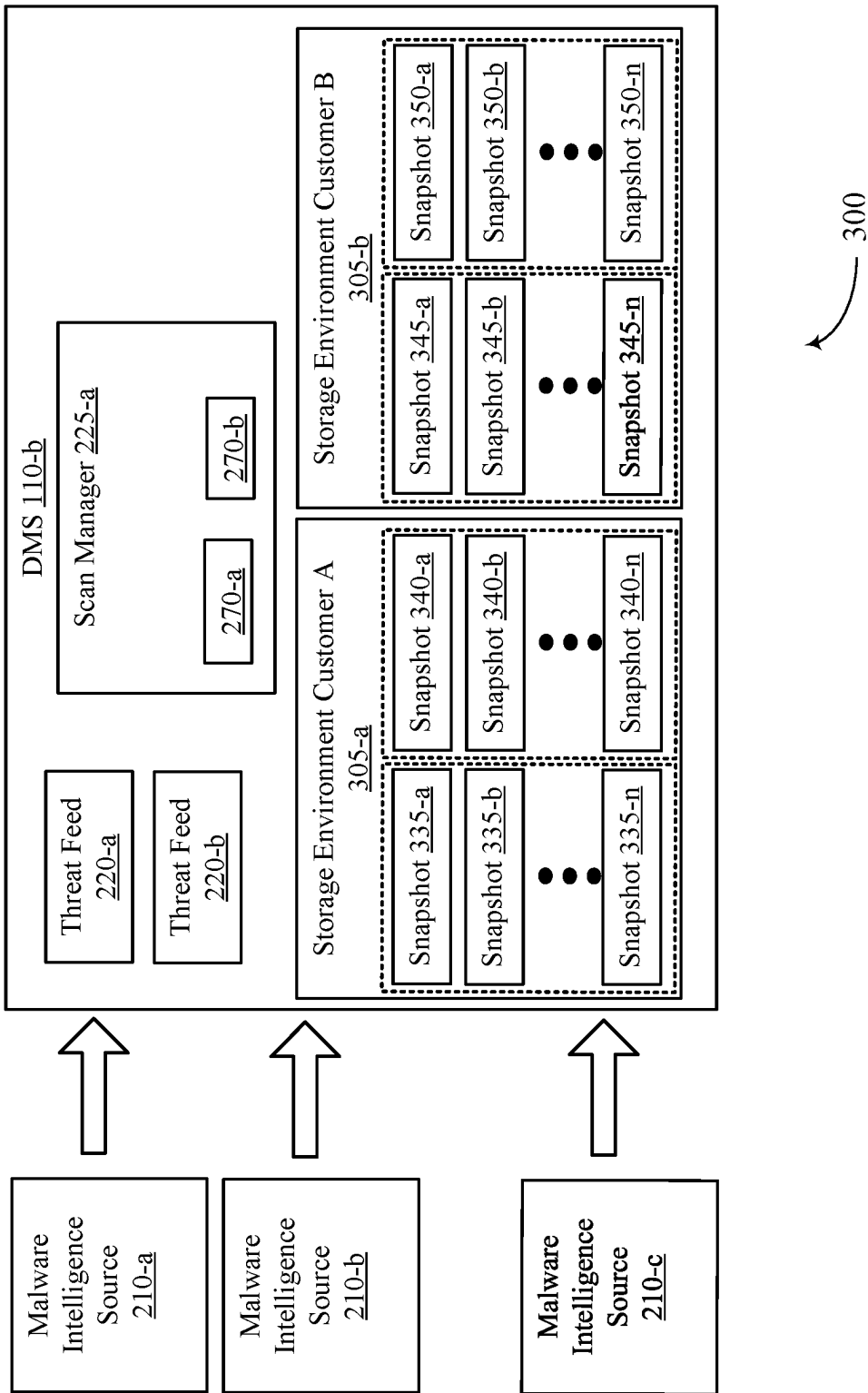


FIG. 3

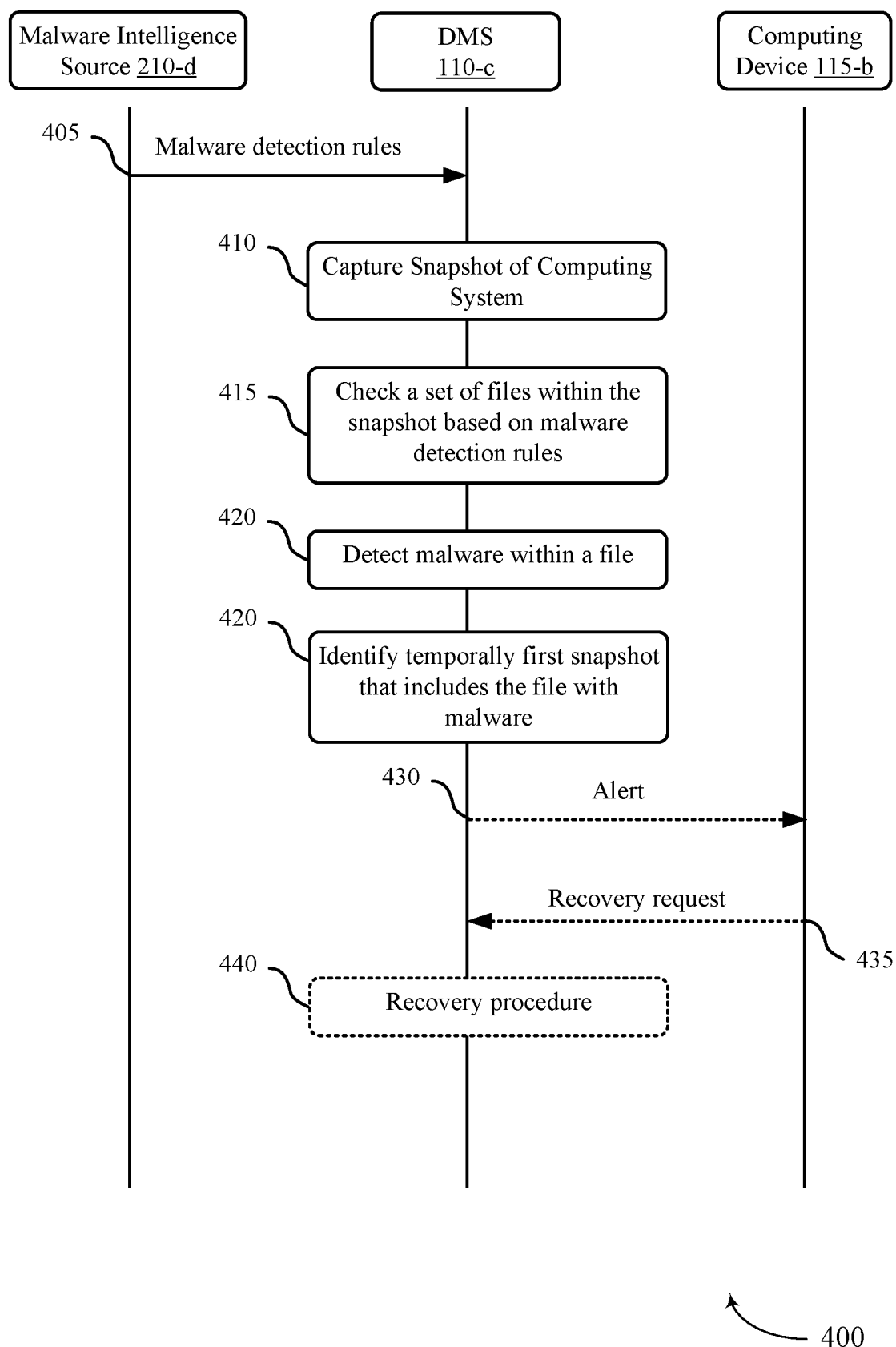


FIG. 4

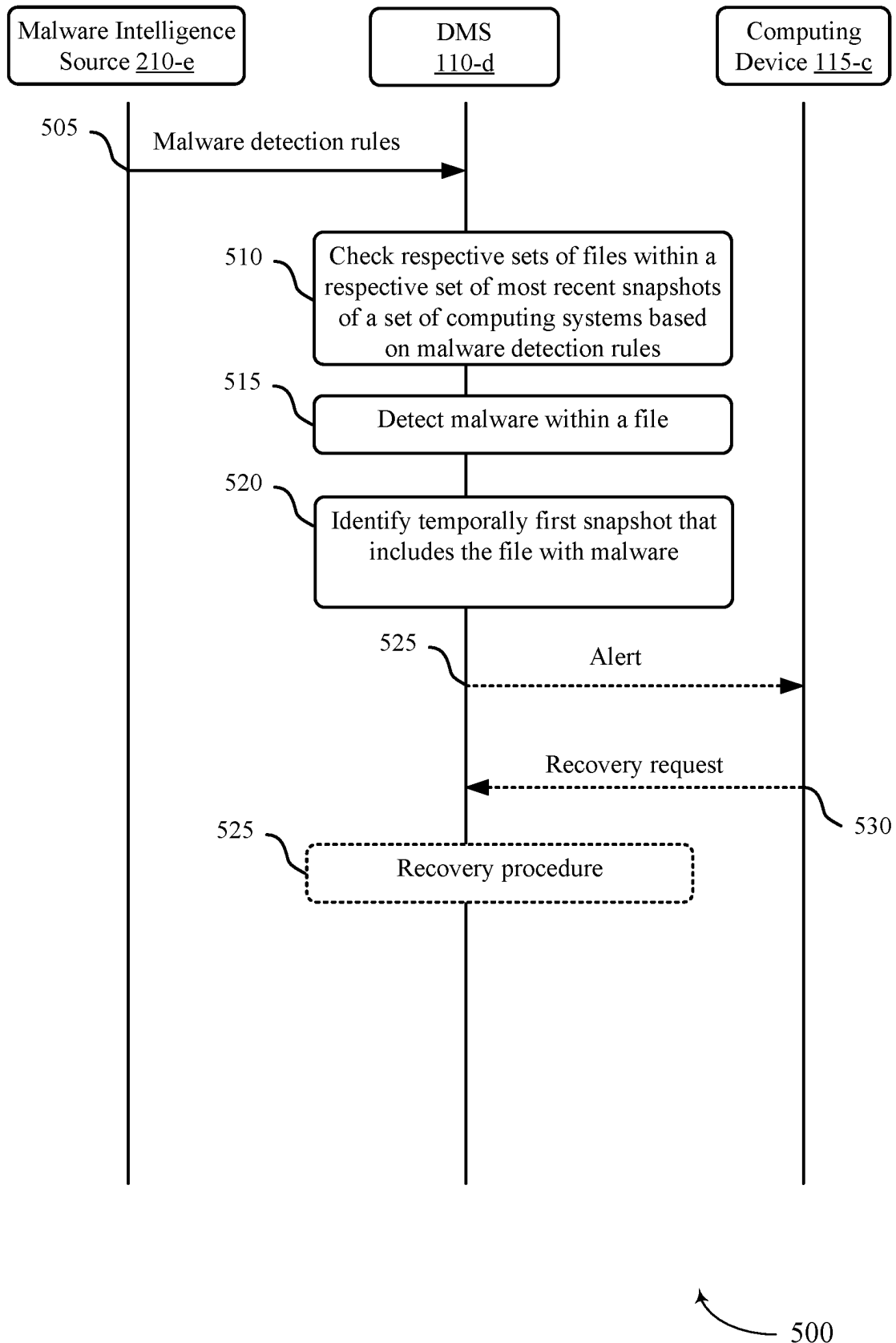


FIG. 5

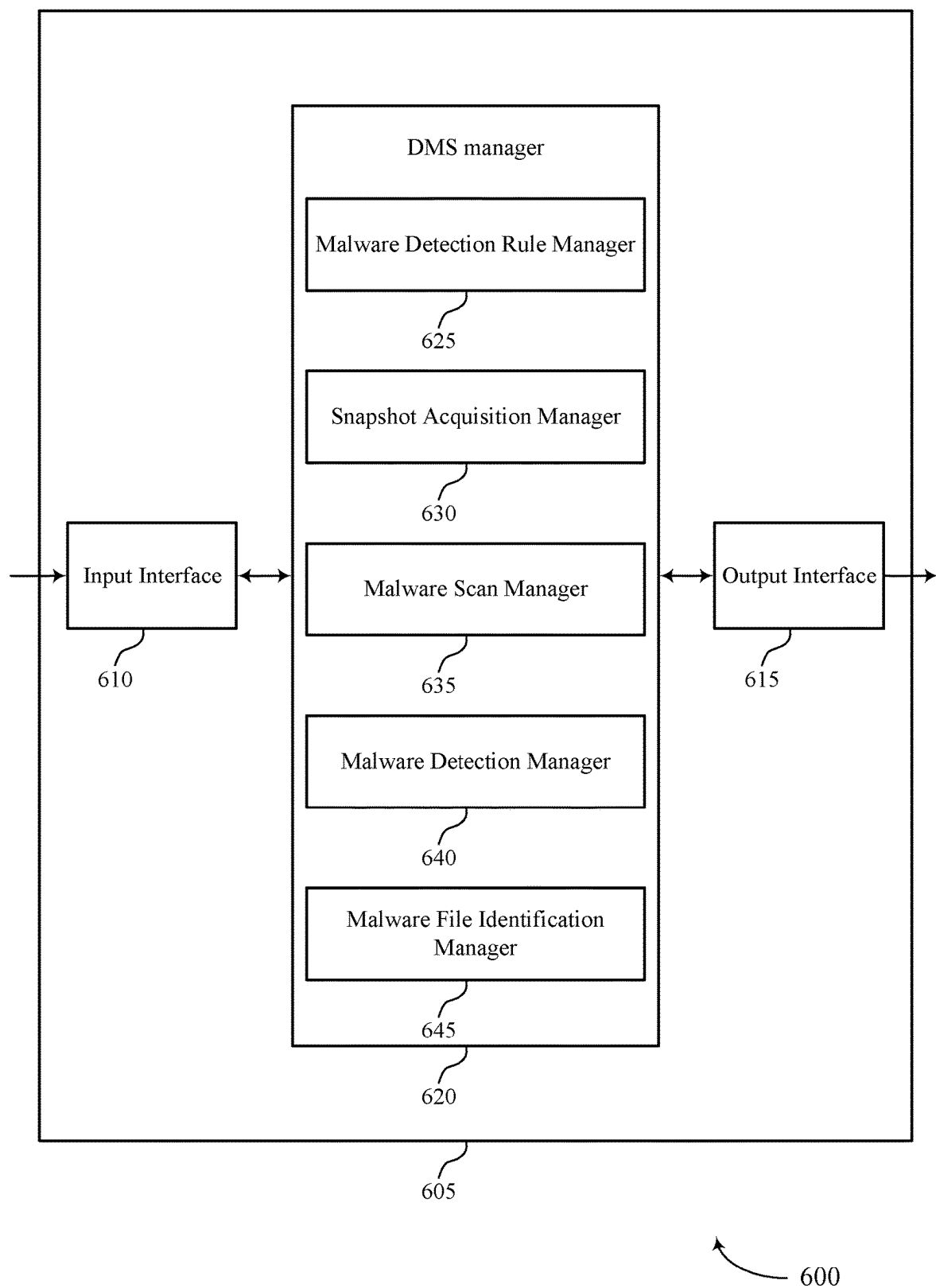


FIG. 6

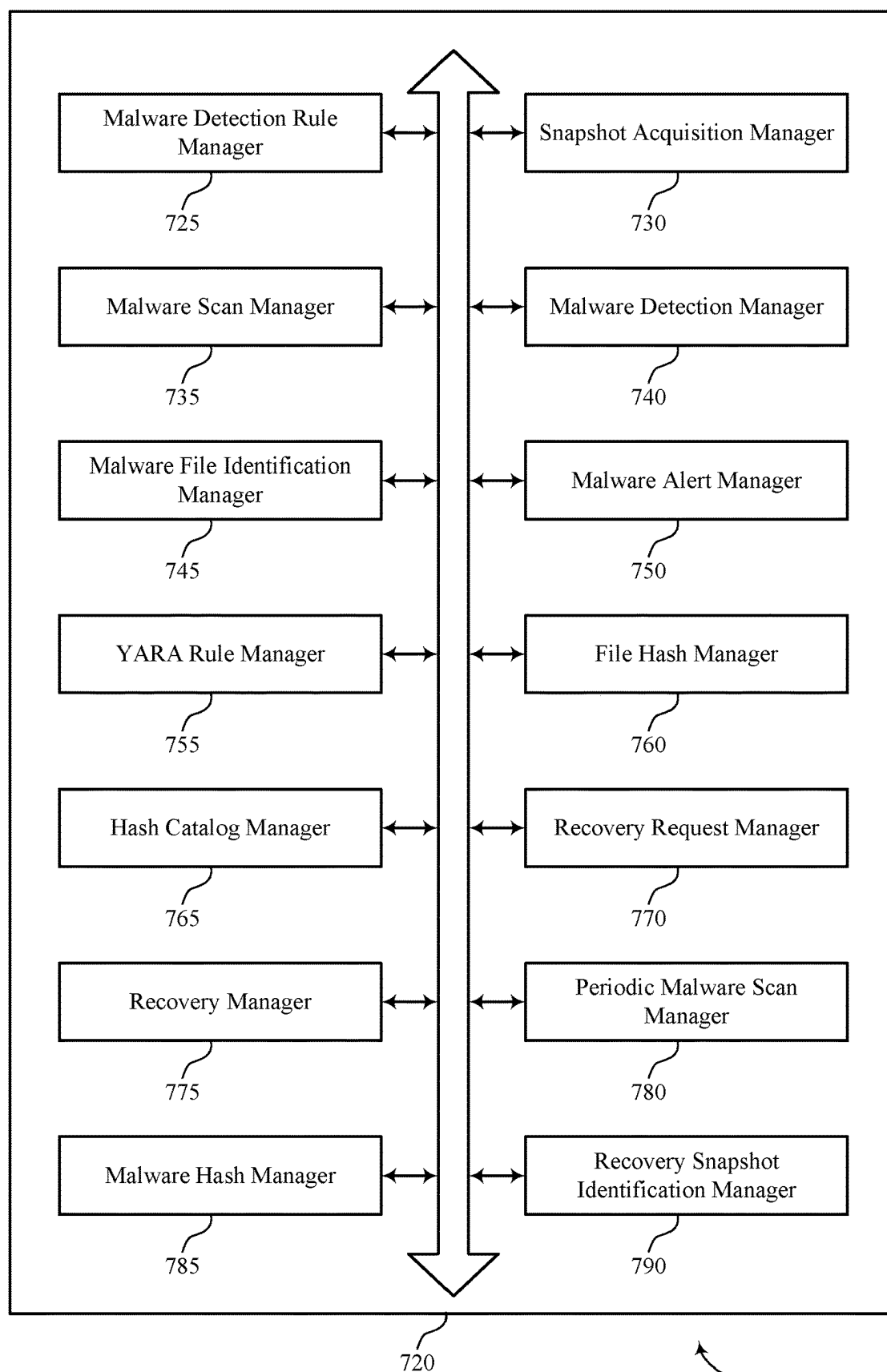


FIG. 7

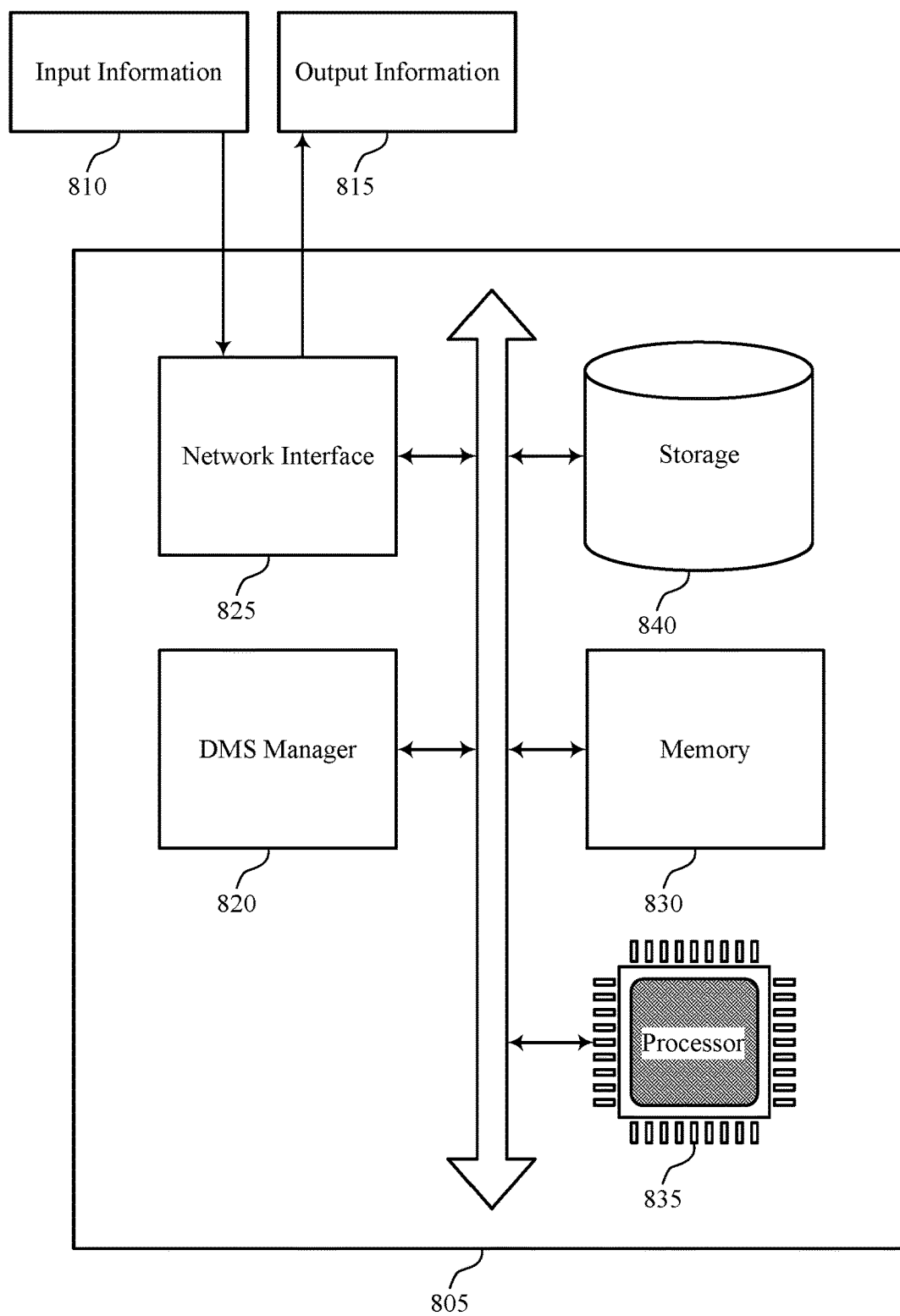
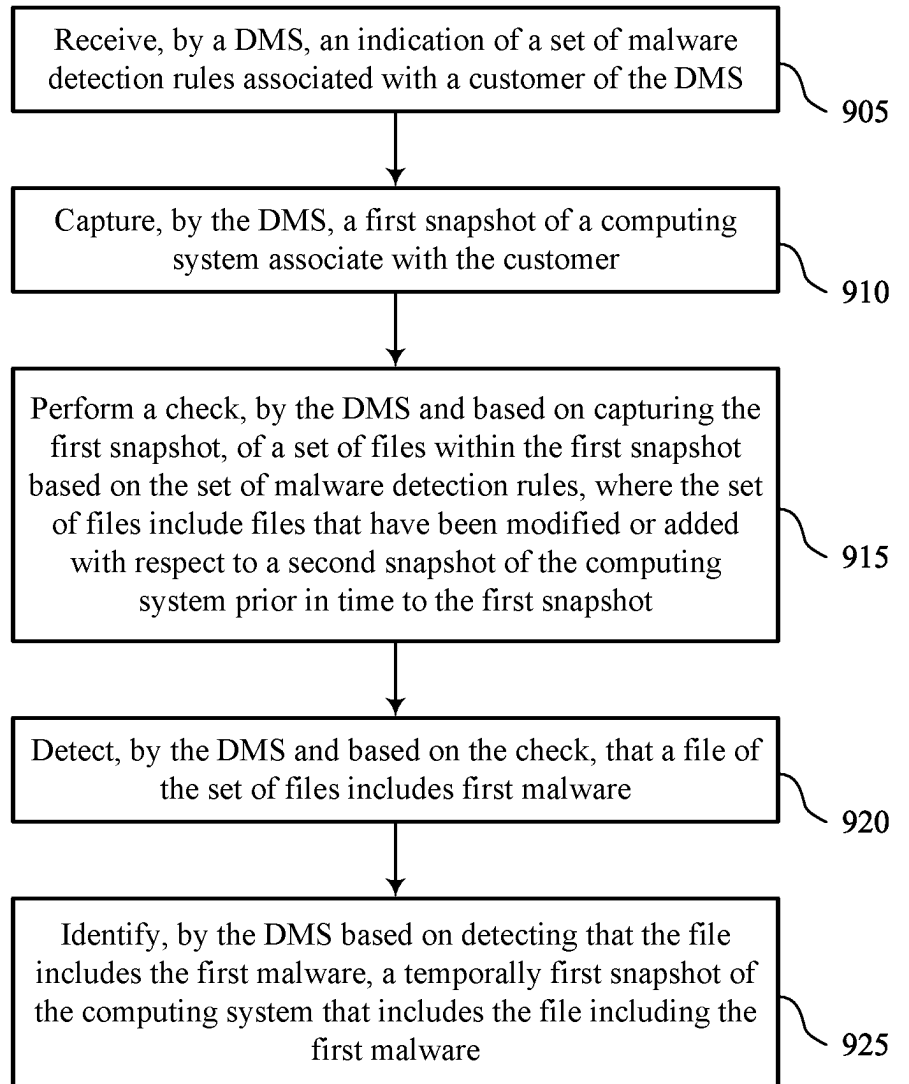


FIG. 8



900

FIG. 9

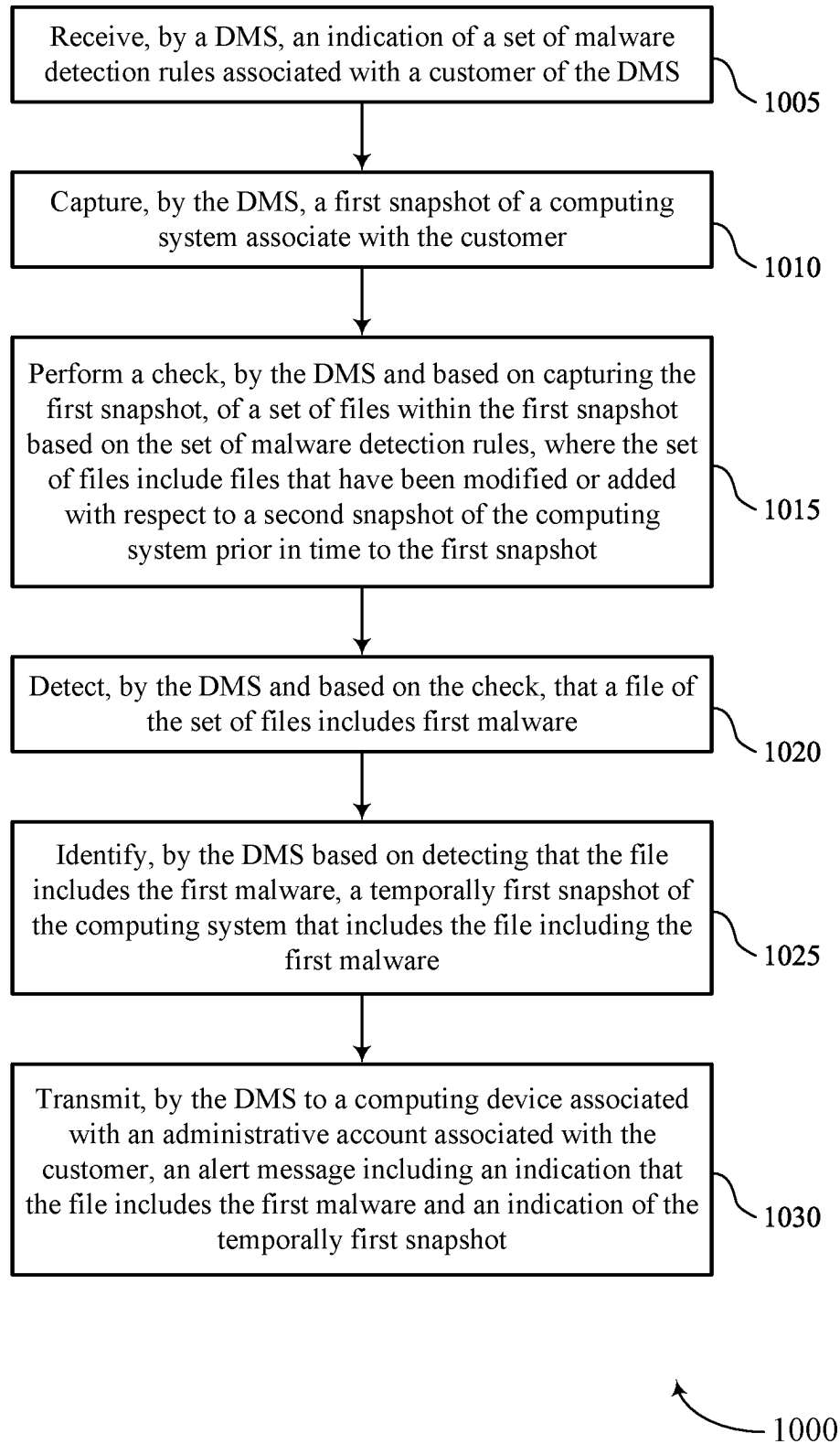


FIG. 10

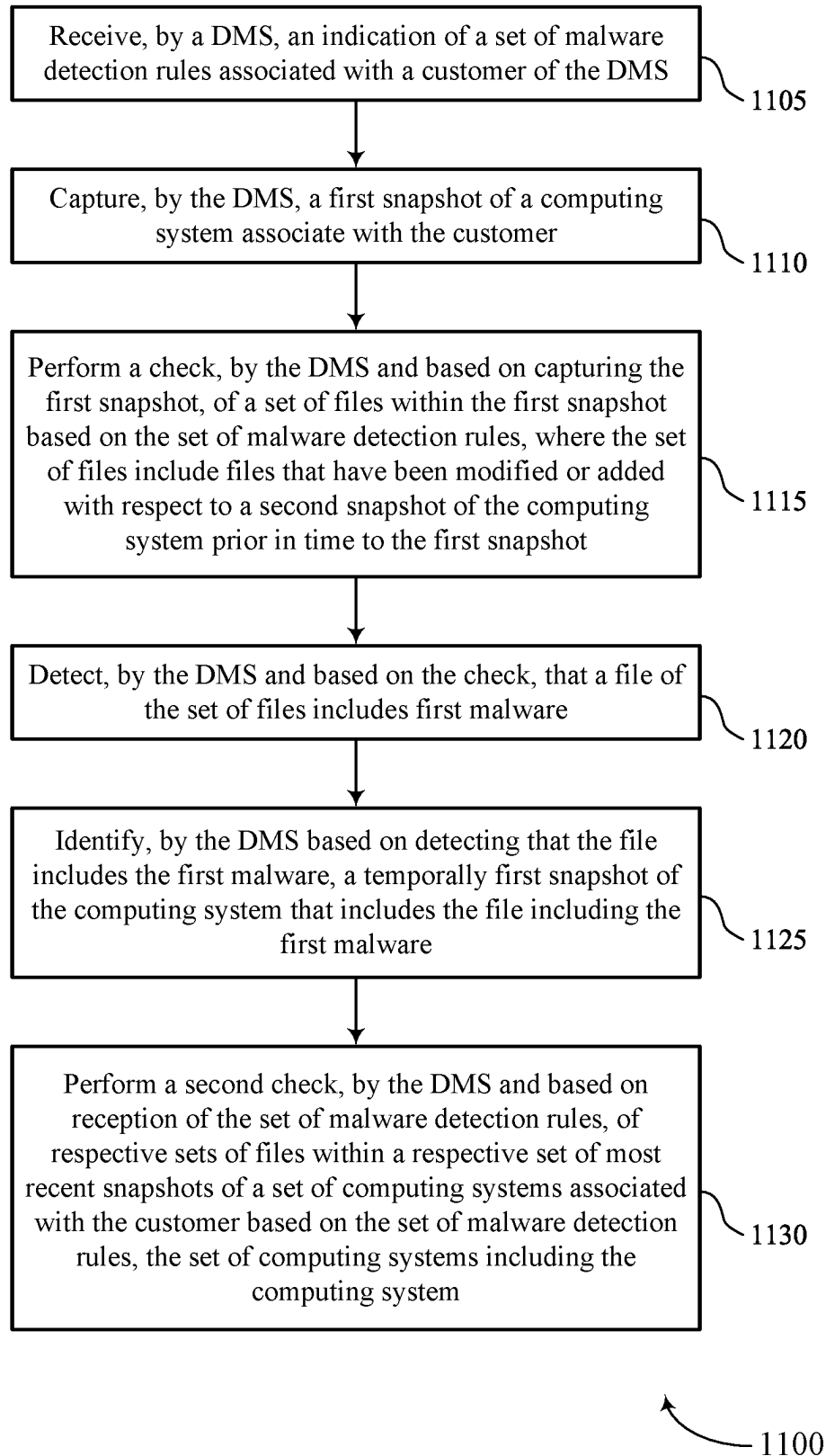


FIG. 11

MALWARE MONITORING AND DETECTION

FIELD OF TECHNOLOGY

[0001] The present disclosure relates generally to data management, including techniques for malware monitoring and detection.

BACKGROUND

[0002] A data management system (DMS) may be employed to manage data associated with one or more computing systems. The data may be generated, stored, or otherwise used by the one or more computing systems, examples of which may include servers, databases, virtual machines, cloud computing systems, file systems (e.g., network-attached storage (NAS) systems), or other data storage or processing systems. The DMS may provide data backup, data recovery, data classification, or other types of data management services for data of the one or more computing systems. Improved data management may offer improved performance with respect to reliability, speed, efficiency, scalability, security, or ease-of-use, among other possible aspects of performance.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIGS. 1, 2, and 3 illustrate examples of computing environments that supports malware monitoring and detection in accordance with aspects of the present disclosure.

[0004] FIG. 4 shows an example of a process flow that supports malware monitoring and detection in accordance with aspects of the present disclosure.

[0005] FIG. 5 shows an example of a process flow that supports malware monitoring and detection in accordance with aspects of the present disclosure.

[0006] FIG. 6 shows a block diagram of an apparatus that supports malware monitoring and detection in accordance with aspects of the present disclosure.

[0007] FIG. 7 shows a block diagram of a DMS manager that supports malware monitoring and detection in accordance with aspects of the present disclosure.

[0008] FIG. 8 shows a diagram of a system including a device that supports malware monitoring and detection in accordance with aspects of the present disclosure.

[0009] FIGS. 9 through 11 show flowcharts illustrating methods that support malware monitoring and detection in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

[0010] A data management system (DMS) may include various nodes, clusters, and sub-systems that provide backup and recovery services for customer computing systems. Backup processes may involve capturing snapshots of customer computing systems. The DMS may generate metadata when capturing snapshots, for example, in the form of differential filesystem metadata (diff FMD) files which contain information corresponding to files that have been created, deleted, modified, or otherwise changed since a prior snapshot. Malware may infect computing systems, and new strains of malware constantly may be emerging. Customers may have large amounts of data, which may create challenges with respect to maintaining up to date scans for malware. Further, some malware scanning techniques, such as endpoint protection, do not maintain historical records of

computing systems and therefore do not provide a safe recovery point upon detection of malware.

[0011] Aspects of this disclosure relate to maintaining a malware threat feed at the DMS that includes updated malware intelligence. For example, the malware intelligence may involve YARA rules and known hashes indicative of malware. The DMS may use the malware intelligence to scan each new snapshot of a customer computing system to determine if the new snapshot contains malware (e.g., the DMS may perform incremental scans on each new snapshot). When checking a new snapshot, the DMS may check the files that were added or modified in the new snapshot (e.g., based on the diff FMD file) to determine whether the malware was introduced into the customer computing system since the snapshot immediately prior to the new snapshot or to determine the first snapshot into which the malware was introduced. Further, when new malware intelligence is received at the DMS, the DMS may check the files within the most recent snapshot of each computing system for a customer to determine whether any computing system of the customer contains malware (e.g., the DMS may perform a full scan for the customer based on newly acquired malware intelligence). A combination of performance of incremental scans on each new snapshot using a full set of malware intelligence maintained at a threat feed and performance of full scans based on newly acquired malware intelligence may ensure that each file backed up by a DMS for a customer is scanned based on up to date malware intelligence information.

[0012] In some examples, when malware is detected in a file within a snapshot—either based on checking a newly captured snapshot (e.g., based on an incremental scan) or when checking snapshots based on new malware intelligence (e.g., based on a full scan)—the DMS may determine the temporally first snapshot which contained the corrupted file. Accordingly, the DMS may identify the snapshot immediately prior to that temporally first snapshot as a safe recovery point. The DMS may provide backup and recovery services for multiple customers, and each customer may have their own malware intelligence feed. Further, the DMS may maintain a malware intelligence feed that is applicable to multiple or all customers.

[0013] FIG. 1 illustrates an example of a computing environment 100 that supports malware monitoring and detection in accordance with aspects of the present disclosure. The computing environment 100 may include a computing system 105, a DMS 110, and one or more computing devices 115, which may be in communication with one another via a network 120. The computing system 105 may generate, store, process, modify, or otherwise use associated data, and the DMS 110 may provide one or more data management services for the computing system 105. For example, the DMS 110 may provide a data backup service, a data recovery service, a data classification service, a data transfer or replication service, one or more other data management services, or any combination thereof for data associated with the computing system 105.

[0014] The network 120 may allow the one or more computing devices 115, the computing system 105, and the DMS 110 to communicate (e.g., exchange information) with one another. The network 120 may include aspects of one or more wired networks (e.g., the Internet), one or more wireless networks (e.g., cellular networks), or any combination thereof. The network 120 may include aspects of one

or more public networks or private networks, as well as secured or unsecured networks, or any combination thereof. The network 120 also may include any quantity of communications links and any quantity of hubs, bridges, routers, switches, ports or other physical or logical network components.

[0015] A computing device 115 may be used to input information to or receive information from the computing system 105, the DMS 110, or both. For example, a user of the computing device 115 may provide user inputs via the computing device 115, which may result in commands, data, or any combination thereof being communicated via the network 120 to the computing system 105, the DMS 110, or both. Additionally, or alternatively, a computing device 115 may output (e.g., display) data or other information received from the computing system 105, the DMS 110, or both. A user of a computing device 115 may, for example, use the computing device 115 to interact with one or more user interfaces (e.g., graphical user interfaces (GUIs)) to operate or otherwise interact with the computing system 105, the DMS 110, or both. Though one computing device 115 is shown in FIG. 1, it is to be understood that the computing environment 100 may include any quantity of computing devices 115.

[0016] A computing device 115 may be a stationary device (e.g., a desktop computer or access point) or a mobile device (e.g., a laptop computer, tablet computer, or cellular phone). In some examples, a computing device 115 may be a commercial computing device, such as a server or collection of servers. And in some examples, a computing device 115 may be a virtual device (e.g., a virtual machine). Though shown as a separate device in the example computing environment of FIG. 1, it is to be understood that in some cases a computing device 115 may be included in (e.g., may be a component of) the computing system 105 or the DMS 110.

[0017] The computing system 105 may include one or more servers 125 and may provide (e.g., to the one or more computing devices 115) local or remote access to applications, databases, or files stored within the computing system 105. The computing system 105 may further include one or more data storage devices 130. Though one server 125 and one data storage device 130 are shown in FIG. 1, it is to be understood that the computing system 105 may include any quantity of servers 125 and any quantity of data storage devices 130, which may be in communication with one another and collectively perform one or more functions ascribed herein to the server 125 and data storage device 130.

[0018] A data storage device 130 may include one or more hardware storage devices operable to store data, such as one or more hard disk drives (HDDs), magnetic tape drives, solid-state drives (SSDs), storage area network (SAN) storage devices, or network-attached storage (NAS) devices. In some cases, a data storage device 130 may comprise a tiered data storage infrastructure (or a portion of a tiered data storage infrastructure). A tiered data storage infrastructure may allow for the movement of data across different tiers of the data storage infrastructure between higher-cost, higher-performance storage devices (e.g., SSDs and HDDs) and relatively lower-cost, lower-performance storage devices (e.g., magnetic tape drives). In some examples, a data storage device 130 may be a database (e.g., a relational

database), and a server 125 may host (e.g., provide a database management system for) the database.

[0019] A server 125 may allow a client (e.g., a computing device 115) to download information or files (e.g., executable, text, application, audio, image, or video files) from the computing system 105, to upload such information or files to the computing system 105, or to perform a search query related to particular information stored by the computing system 105. In some examples, a server 125 may act as an application server or a file server. In general, a server 125 may refer to one or more hardware devices that act as the host in a client-server relationship or a software process that shares a resource with or performs work for one or more clients.

[0020] A server 125 may include a network interface 140, processor 145, memory 150, disk 155, and computing system manager 160. The network interface 140 may enable the server 125 to connect to and exchange information via the network 120 (e.g., using one or more network protocols). The network interface 140 may include one or more wireless network interfaces, one or more wired network interfaces, or any combination thereof. The processor 145 may execute computer-readable instructions stored in the memory 150 in order to cause the server 125 to perform functions ascribed herein to the server 125. The processor 145 may include one or more processing units, such as one or more central processing units (CPUs), one or more graphics processing units (GPUs), or any combination thereof. The memory 150 may comprise one or more types of memory (e.g., random access memory (RAM), static random access memory (SRAM), dynamic random access memory (DRAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), Flash, etc.). Disk 155 may include one or more HDDs, one or more SSDs, or any combination thereof. Memory 150 and disk 155 may comprise hardware storage devices. The computing system manager 160 may manage the computing system 105 or aspects thereof (e.g., based on instructions stored in the memory 150 and executed by the processor 145) to perform functions ascribed herein to the computing system 105. In some examples, the network interface 140, processor 145, memory 150, and disk 155 may be included in a hardware layer of a server 125, and the computing system manager 160 may be included in a software layer of the server 125. In some cases, the computing system manager 160 may be distributed across (e.g., implemented by) multiple servers 125 within the computing system 105.

[0021] In some examples, the computing system 105 or aspects thereof may be implemented within one or more cloud computing environments, which may alternatively be referred to as cloud environments. Cloud computing may refer to Internet-based computing, wherein shared resources, software, and/or information may be provided to one or more computing devices on-demand via the Internet. A cloud environment may be provided by a cloud platform, where the cloud platform may include physical hardware components (e.g., servers) and software components (e.g., operating system) that implement the cloud environment. A cloud environment may implement the computing system 105 or aspects thereof through Software-as-a-Service (SaaS) or Infrastructure-as-a-Service (IaaS) services provided by the cloud environment. SaaS may refer to a software distribution model in which applications are hosted by a service provider and made available to one or more client devices

over a network (e.g., to one or more computing devices **115** over the network **120**). IaaS may refer to a service in which physical computing resources are used to instantiate one or more virtual machines, the resources of which are made available to one or more client devices over a network (e.g., to one or more computing devices **115** over the network **120**).

[0022] In some examples, the computing system **105** or aspects thereof may implement or be implemented by one or more virtual machines. The one or more virtual machines may run various applications, such as a database server, an application server, or a web server. For example, a server **125** may be used to host (e.g., create, manage) one or more virtual machines, and the computing system manager **160** may manage a virtualized infrastructure within the computing system **105** and perform management operations associated with the virtualized infrastructure. The computing system manager **160** may manage the provisioning of virtual machines running within the virtualized infrastructure and provide an interface to a computing device **115** interacting with the virtualized infrastructure. For example, the computing system manager **160** may be or include a hypervisor and may perform various virtual machine-related tasks, such as cloning virtual machines, creating new virtual machines, monitoring the state of virtual machines, moving virtual machines between physical hosts for load balancing purposes, and facilitating backups of virtual machines. In some examples, the virtual machines, the hypervisor, or both, may virtualize and make available resources of the disk **155**, the memory, the processor **145**, the network interface **140**, the data storage device **130**, or any combination thereof in support of running the various applications. Storage resources (e.g., the disk **155**, the memory **150**, or the data storage device **130**) that are virtualized may be accessed by applications as a virtual disk.

[0023] The DMS **110** may provide one or more data management services for data associated with the computing system **105** and may include DMS manager **190** and any quantity of storage nodes **185**. The DMS manager **190** may manage operation of the DMS **110**, including the storage nodes **185**. Though illustrated as a separate entity within the DMS **110**, the DMS manager **190** may in some cases be implemented (e.g., as a software application) by one or more of the storage nodes **185**. In some examples, the storage nodes **185** may be included in a hardware layer of the DMS **110**, and the DMS manager **190** may be included in a software layer of the DMS **110**. In the example illustrated in FIG. 1, the DMS **110** is separate from the computing system **105** but in communication with the computing system **105** via the network **120**. It is to be understood, however, that in some examples at least some aspects of the DMS **110** may be located within computing system **105**. For example, one or more servers **125**, one or more data storage devices **130**, and at least some aspects of the DMS **110** may be implemented within the same cloud environment or within the same data center.

[0024] Storage nodes **185** of the DMS **110** may include respective network interfaces **165**, processors **170**, memories **175**, and disks **180**. The network interfaces **165** may enable the storage nodes **185** to connect to one another, to the network **120**, or both. A network interface **165** may include one or more wireless network interfaces, one or more wired network interfaces, or any combination thereof. The processor **170** of a storage node **185** may execute

computer-readable instructions stored in the memory **175** of the storage node **185** in order to cause the storage node **185** to perform processes described herein as performed by the storage node **185**. A processor **170** may include one or more processing units, such as one or more CPUs, one or more GPUs, or any combination thereof. The memory **150** may comprise one or more types of memory (e.g., RAM, SRAM, DRAM, ROM, EEPROM, Flash, etc.). A disk **180** may include one or more HDDs, one or more SSDs, or any combination thereof. Memories **175** and disks **180** may comprise hardware storage devices. Collectively, the storage nodes **185** may in some cases be referred to as a storage cluster or as a cluster of storage nodes **185**.

[0025] The DMS **110** may provide a backup and recovery service for the computing system **105**. For example, the DMS **110** may manage the extraction and storage of snapshots **135** associated with different point-in-time versions of one or more target computing objects within the computing system **105**. A snapshot **135** of a computing object (e.g., a virtual machine, a database, a filesystem, a virtual disk, a virtual desktop, or other type of computing system or storage system) may be a file (or set of files) that represents a state of the computing object (e.g., the data thereof) as of a particular point in time. A snapshot **135** may also be used to restore (e.g., recover) the corresponding computing object as of the particular point in time corresponding to the snapshot **135**. A computing object of which a snapshot **135** may be generated may be referred to as snappable. Snapshots **135** may be generated at different times (e.g., periodically or on some other scheduled or configured basis) in order to represent the state of the computing system **105** or aspects thereof as of those different times. In some examples, a snapshot **135** may include metadata that defines a state of the computing object as of a particular point in time. For example, a snapshot **135** may include metadata associated with (e.g., that defines a state of) some or all data blocks included in (e.g., stored by or otherwise included in) the computing object. Snapshots **135** (e.g., collectively) may capture changes in the data blocks over time. Snapshots **135** generated for the target computing objects within the computing system **105** may be stored in one or more storage locations (e.g., the disk **155**, memory **150**, the data storage device **130**) of the computing system **105**, in the alternative or in addition to being stored within the DMS **110**, as described below.

[0026] To obtain a snapshot **135** of a target computing object associated with the computing system **105** (e.g., of the entirety of the computing system **105** or some portion thereof, such as one or more databases, virtual machines, or filesystems within the computing system **105**), the DMS manager **190** may transmit a snapshot request to the computing system manager **160**. In response to the snapshot request, the computing system manager **160** may set the target computing object into a frozen state (e.g., a read-only state). Setting the target computing object into a frozen state may allow a point-in-time snapshot **135** of the target computing object to be stored or transferred.

[0027] In some examples, the computing system **105** may generate the snapshot **135** based on the frozen state of the computing object. For example, the computing system **105** may execute an agent of the DMS **110** (e.g., the agent may be software installed at and executed by one or more servers **125**), and the agent may cause the computing system **105** to generate the snapshot **135** and transfer the snapshot **135** to

the DMS 110 in response to the request from the DMS 110. In some examples, the computing system manager 160 may cause the computing system 105 to transfer, to the DMS 110, data that represents the frozen state of the target computing object, and the DMS 110 may generate a snapshot 135 of the target computing object based on the corresponding data received from the computing system 105.

[0028] Once the DMS 110 receives, generates, or otherwise obtains a snapshot 135, the DMS 110 may store the snapshot 135 at one or more of the storage nodes 185. The DMS 110 may store a snapshot 135 at multiple storage nodes 185, for example, for improved reliability. Additionally, or alternatively, snapshots 135 may be stored in some other location connected with the network 120. For example, the DMS 110 may store more recent snapshots 135 at the storage nodes 185, and the DMS 110 may transfer less recent snapshots 135 via the network 120 to a cloud environment (which may include or be separate from the computing system 105) for storage at the cloud environment, a magnetic tape storage device, or another storage system separate from the DMS 110.

[0029] Updates made to a target computing object that has been set into a frozen state may be written by the computing system 105 to a separate file (e.g., an update file) or other entity within the computing system 105 while the target computing object is in the frozen state. After the snapshot 135 (or associated data) of the target computing object has been transferred to the DMS 110, the computing system manager 160 may release the target computing object from the frozen state, and any corresponding updates written to the separate file or other entity may be merged into the target computing object.

[0030] In response to a restore command (e.g., from a computing device 115 or the computing system 105), the DMS 110 may restore a target version (e.g., corresponding to a particular point in time) of a computing object based on a corresponding snapshot 135 of the computing object. In some examples, the corresponding snapshot 135 may be used to restore the target version based on data of the computing object as stored at the computing system 105 (e.g., based on information included in the corresponding snapshot 135 and other information stored at the computing system 105, the computing object may be restored to its state as of the particular point in time). Additionally, or alternatively, the corresponding snapshot 135 may be used to restore the data of the target version based on data of the computing object as included in one or more backup copies of the computing object (e.g., file-level backup copies or image-level backup copies). Such backup copies of the computing object may be generated in conjunction with or according to a separate schedule than the snapshots 135. For example, the target version of the computing object may be restored based on the information in a snapshot 135 and based on information included in a backup copy of the target object generated prior to the time corresponding to the target version. Backup copies of the computing object may be stored at the DMS 110 (e.g., in the storage nodes 185) or in some other location connected with the network 120 (e.g., in a cloud environment, which in some cases may be separate from the computing system 105).

[0031] In some examples, the DMS 110 may restore the target version of the computing object and transfer the data of the restored computing object to the computing system 105. And in some examples, the DMS 110 may transfer one

or more snapshots 135 to the computing system 105, and restoration of the target version of the computing object may occur at the computing system 105 (e.g., as managed by an agent of the DMS 110, where the agent may be installed and operate at the computing system 105).

[0032] In response to a mount command (e.g., from a computing device 115 or the computing system 105), the DMS 110 may instantiate data associated with a point-in-time version of a computing object based on a snapshot 135 corresponding to the computing object (e.g., along with data included in a backup copy of the computing object) and the point-in-time. The DMS 110 may then allow the computing system 105 to read or modify the instantiated data (e.g., without transferring the instantiated data to the computing system). In some examples, the DMS 110 may instantiate (e.g., virtually mount) some or all of the data associated with the point-in-time version of the computing object for access by the computing system 105, the DMS 110, or the computing device 115.

[0033] In some examples, the DMS 110 may store different types of snapshots 135, including for the same computing object. For example, the DMS 110 may store both base snapshots 135 and incremental snapshots 135. A base snapshot 135 may represent the entirety of the state of the corresponding computing object as of a point in time corresponding to the base snapshot 135. An incremental snapshot 135 may represent the changes to the state—which may be referred to as the delta—of the corresponding computing object that have occurred between an earlier or later point in time corresponding to another snapshot 135 (e.g., another base snapshot 135 or incremental snapshot 135) of the computing object and the incremental snapshot 135. In some cases, some incremental snapshots 135 may be forward-incremental snapshots 135 and other incremental snapshots 135 may be reverse-incremental snapshots 135. To generate a full snapshot 135 of a computing object using a forward-incremental snapshot 135, the information of the forward-incremental snapshot 135 may be combined with (e.g., applied to) the information of an earlier base snapshot 135 of the computing object along with the information of any intervening forward-incremental snapshots 135, where the earlier base snapshot 135 may include a base snapshot 135 and one or more reverse-incremental or forward-incremental snapshots 135. To generate a full snapshot 135 of a computing object using a reverse-incremental snapshot 135, the information of the reverse-incremental snapshot 135 may be combined with (e.g., applied to) the information of a later base snapshot 135 of the computing object along with the information of any intervening reverse-incremental snapshots 135.

[0034] In some examples, the DMS 110 may provide a data classification service, a malware detection service, a data transfer or replication service, backup verification service, or any combination thereof, among other possible data management services for data associated with the computing system 105. For example, the DMS 110 may analyze data included in one or more computing objects of the computing system 105, metadata for one or more computing objects of the computing system 105, or any combination thereof, and based on such analysis, the DMS 110 may identify locations within the computing system 105 that include data of one or more target data types (e.g., sensitive data, such as data subject to privacy regulations or otherwise of particular interest) and output related information (e.g., for display to

a user via a computing device **115**). Additionally, or alternatively, the DMS **110** may detect whether aspects of the computing system **105** have been impacted by malware (e.g., ransomware). Additionally, or alternatively, the DMS **110** may relocate data or create copies of data based on using one or more snapshots **135** to restore the associated computing object within its original location or at a new location (e.g., a new location within a different computing system **105**). Additionally, or alternatively, the DMS **110** may analyze backup data to ensure that the underlying data (e.g., user data or metadata) has not been corrupted. The DMS **110** may perform such data classification, malware detection, data transfer or replication, or backup verification, for example, based on data included in snapshots **135** or backup copies of the computing system **105**, rather than live contents of the computing system **105**, which may beneficially avoid adversely affecting (e.g., infecting, loading, etc.) the computing system **105**.

[0035] In some examples, the DMS **110**, and in particular the DMS manager **190**, may be referred to as a control plane. The control plane may manage tasks, such as storing data management data or performing restorations, among other possible examples. The control plane may be common to multiple customers or tenants of the DMS **110**. For example, the computing system **105** may be associated with a first customer or tenant of the DMS **110**, and the DMS **110** may similarly provide data management services for one or more other computing systems associated with one or more additional customers or tenants. In some examples, the control plane may be configured to manage the transfer of data management data (e.g., snapshots **135** associated with the computing system **105**) to a cloud environment **195** (e.g., Microsoft Azure or Amazon Web Services). In addition, or as an alternative, to being configured to manage the transfer of data management data to the cloud environment **195**, the control plane may be configured to transfer metadata for the data management data to the cloud environment **195**. The metadata may be configured to facilitate storage of the stored data management data, the management of the stored management data, the processing of the stored management data, the restoration of the stored data management data, and the like.

[0036] Each customer or tenant of the DMS **110** may have a private data plane, where a data plane may include a location at which customer or tenant data is stored. For example, each private data plane for each customer or tenant may include a node cluster **196** across which data (e.g., data management data, metadata for data management data, etc.) for a customer or tenant is stored. Each node cluster **196** may include a node controller **197** which manages the nodes **198** of the node cluster **196**. As an example, a node cluster **196** for one tenant or customer may be hosted on Microsoft Azure, and another node cluster **196** may be hosted on Amazon Web Services. In another example, multiple separate node clusters **196** for multiple different customers or tenants may be hosted on Microsoft Azure. Separating each customer or tenant's data into separate node clusters **196** provides fault isolation for the different customers or tenants and provides security by limiting access to data for each customer or tenant.

[0037] The control plane (e.g., the DMS **110**, and specifically the DMS manager **190**) manages tasks, such as storing backups or snapshots **135** or performing restorations, across the multiple node clusters **196**. For example, as described

herein, a node cluster **196-a** may be associated with the first customer or tenant associated with the computing system **105**. The DMS **110** may obtain (e.g., generate or receive) and transfer the snapshots **135** associated with the computing system **105** to the node cluster **196-a** in accordance with a service level agreement for the first customer or tenant associated with the computing system **105**. For example, a service level agreement may define backup and recovery parameters for a customer or tenant such as snapshot generation frequency, which computing objects to backup, where to store the snapshots **135** (e.g., which private data plane), and how long to retain snapshots **135**. As described herein, the control plane may provide data management services for another computing system associated with another customer or tenant. For example, the control plane may generate and transfer snapshots **135** for another computing system associated with another customer or tenant to the node cluster **196-n** in accordance with the service level agreement for the other customer or tenant.

[0038] To manage tasks, such as storing backups or snapshots **135** or performing restorations, across the multiple node clusters **196**, the control plane (e.g., the DMS manager **190**) may communicate with the node controllers **197** for the various node clusters via the network **120**. For example, the control plane may exchange communications for backup and recovery tasks with the node controllers **197** in the form of transmission control protocol (TCP) packets via the network **120**.

[0039] The DMS **110** may generate metadata when capturing snapshots **135**, for example, in the form of differential filesystem metadata (diff FMD) files which contain information corresponding to files that have been created, deleted, modified, or otherwise changed since a prior snapshot **135**.

[0040] The DMS **110** may maintain a malware threat feed (e.g., via the network **120**) that includes updated malware intelligence. For example, the DMS **110** may receive malware intelligence from malware intelligence subscriptions. For example, a customer of the DMS **110** may subscribe to one or more malware intelligence services, and/or the DMS **110** may subscribe to one or more malware intelligence services. For example, the malware intelligence may involve YARA rules and known hashes indicative of malware. The DMS **110** may use the malware intelligence to scan each new snapshot **135** of the computing system **105** to determine if the new snapshot **135** contains malware. When checking a new snapshot **135**, the DMS **110** may check the files that were added or modified in the new snapshot **135** (e.g., based on the diff FMD file) against the malware intelligence maintained at a threat feed at the DMS **110** to determine whether the malware was introduced into the customer computing system since the snapshot **135** immediately prior to the new snapshot **135** or to determine the first snapshot **135** into which the malware was introduced.

[0041] In some examples, when new malware intelligence is received at the DMS **110**, or periodically in accordance with a service level agreement (SLA) with a customer, the DMS **110** may check the files within the most recent snapshot of each computing system for a customer to determine whether any computing system of the customer contains malware. When malware is detected in a file within a snapshot **135**—either based on checking a newly captured snapshot **135** or when checking snapshots **135** based on new malware intelligence—the DMS **110** may determine the

temporally first snapshot **135** which contained the corrupted file. Accordingly, the DMS may identify the snapshot **135** immediately prior to that temporally first snapshot **135** as a safe recovery point. The DMS **110** may provide backup and recovery services for multiple customers, and each customer may have their own malware intelligence feed. Further, the DMS **110** may maintain a malware intelligence feed that is applicable to multiple or all customers.

[0042] FIG. 2 shows an example of a computing environment **200** that supports malware monitoring and detection in accordance with aspects of the present disclosure. The computing environment **200** may implement one or more aspects of the computing environment **100**. For example, the computing environment **200** includes a DMS **110-a** which may be an example of the DMS **110** as described herein. The computing environment **200** may include a computing device **115-a**, which may be an example of a computing device **115** as described herein. The computing environment **200** may include a computing system **105-a** and a computing system **105-b**, which may be examples of computing systems **105** as described herein.

[0043] The DMS **110-a** may provide backup services for a customer. For example, the DMS **110-a** may capture and store snapshots of computing systems **105** associated with the customer. For example, the DMS **110-a** may capture and store the snapshots **235** of a computing system **105-a** of the customer and snapshots **240** of a computing system **105-b** in a storage environment **205**. In some examples, the storage environment **205** may be one or more storage nodes **185** at the DMS **110-a** as described with reference to FIG. 1. In some examples, the storage environment **205** may be an external storage environment accessible to the DMS **110-a** such as the cloud environment **195** as described with reference to FIG. 1.

[0044] The DMS **110-a** may receive new malware intelligence information **215** (e.g., a set of malware detection rules) from a malware intelligence source **210**. For example, the DMS **110-a** may maintain a malware intelligence subscription that may provide periodic updates or updated malware intelligence as the intelligence is acquired by the malware intelligence subscription. In some examples, in addition to or alternatively to the DMS malware intelligence subscription, the customer may have a subscription to a malware intelligence source to which the DMS **110-a** may have access. The DMS **110-a** may maintain a threat feed **220** which includes the new malware intelligence information **215** received from malware intelligence source **210**. For example, the threat feed **220** may accumulate and/or update malware intelligence information received from the malware intelligence source **210** or from multiple malware intelligence sources. In some examples, the malware intelligence information maintained at the threat feed **220** may include hashes indicative of malware. In some examples, the malware intelligence information maintained at the threat feed **220** may include YARA rules. YARA rules may classify and identify malware samples and may construct descriptions of malware families rooted in textual or binary patterns. YARA rules may be structured around sets of strings, regular expressions, and conditional logic that may be designed to detect specific byte sequences or behaviors in the analyzed malware samples. YARA rules may be used to check files and networks for patterns, scripts and signatures that indicate the presence of malicious software (e.g., malware). In some examples, new YARA rules may be pub-

lished to the DMS **110-a** on a regular basis. In some examples, the DMS **110-a** may generate malware intelligence internally (e.g., may generate malware intelligence at the threat feed) based on malware known to an administrator of the DMS **110-a**.

[0045] The DMS **110-a** may include a scan manager **225** that may use the malware intelligence maintained at the threat feed **220** to scan files in the snapshots (e.g., the snapshots **235** and the snapshots **240**) of the computing systems **105** (e.g., the computing system **105-a** and the computing system **105-b**) associated with the customer. In some examples, the scan manager **225** may include a YARA scan job **245** that may scan files in snapshots against the YARA rules. In some examples, the scan manager **225** may include a hash scan job **250** that may perform hash analysis on hashes of files in the snapshots (e.g., the snapshots **235** and the snapshots **240**) using hash intelligence maintained at the threat feed **220** and provided by a malware intelligence source.

[0046] In some examples, based on capturing a new snapshot **235-n** of the computing system **105-a**, the DMS **110-a** may perform (e.g., the YARA scan job **245** may perform) a YARA scan on files in the new snapshot **235-n** using YARA rules maintained at the threat feed **220** and/or the DMS **110-a** may perform (e.g., the Hash scan job **250** may perform) hash analysis on the new snapshot **235-n** using hash intelligence maintained at the threat feed **220**.

[0047] For example, when the new snapshot **235-n** is captured, the full set of YARA rules maintained at the threat feed **220** may be used to scan the files that are new or modified in the snapshot **235-n** with respect to a prior snapshot **235** of the computing system **105-a**. For example, the prior snapshot **235** may be a previously analyzed snapshot **235** (e.g., a previous snapshot **235** whose new/modified files have been checked against the full set of YARA rules). For example, the prior snapshot **235** may be an immediately prior snapshot **235-n-1** that has been checked against the full set of YARA rules. For example, the DMS **110-a** may identify the files that are new or modified in the snapshot **235-n** with respect to an immediately prior snapshot **235-n-1** based on a diff FMD generated by the DMS **110-a** during the capture process of the snapshot **235-n**. In some examples, the DMS **110-a** may include an incremental scan job manager that may run the YARA scan job **245** for each new snapshot of each computing system associated with the customer. For example, the incremental scan job manager may periodically check for new snapshots associated with the customer added to the storage environment **205** and may run the YARA scan job **245** to check each new snapshot associated with the customer against the latest YARA rules in the threat feed **220**.

[0048] As another example, when the new snapshot **235-n** is captured, the hash scan job may compute hashes of the files in the snapshot **235-n** that are new or modified in the snapshot **235-n** with respect to the immediately prior snapshot **235-n-1** of the computing system **105-a**. The computed hashes of the new or modified files may be compared to a bloom filter **255** representative of the hash intelligence at the threat feed **220**. The bloom filter **255** may be used by the DMS **110-a** to check if new or modified files contain any known malware (e.g., based on the intelligence stored in the threat feed **220**). For example, the bloom filter **255** may enable the DMS **110-a** (e.g., the hash scan job **250**) to compare the hashes of the new or modified files to millions

of known malware strains. The bloom filter **255** may be a space efficient, probabilistic data structure that is used to add elements to a set and check if an element is contained in the set, where the “set” refers to the known indicators of malware (e.g., compromise indicators). If a given computed hash of a modified or added file matches a known indicator of malware based on the bloom filter **255**, the DMS **110-a** (e.g., the hash scan job **250**) may verify the indicated malware in the file against known malware indicators maintained at the threat feed **220**. For example, as the bloom filter **255** may be probabilistic, false positives may occur, and accordingly files that the bloom filter **255** indicates as positive for malware may be checked against a known database of malware to confirm infection.

[0049] If malware is detected in a file of the snapshot **235-a** based on a scan performed by the YARA scan job **245** of the new snapshot **235-n** or based on the hash analysis performed by the hash scan job **250**, an alert **260** may be provided to a computing device **115-a** associated with an administrative account of the customer. For example, the alert **260** may be an email or a message on a user interface at the computing device **115-a** for interfacing with the DMS **110-a** for the customer. The alert **260** may indicate that the file includes malware, and the alert **260** may indicate a temporally first snapshot of the computing system **105-a** that included the file. Accordingly, the customer (e.g., the administrator) may identify a temporally first snapshot of the computing system **105-a** that does not include the file as a safe recovery point for the computing system **105-a**. In some examples, the first snapshot of the computing system **105-a** that included the file may be identified by the DMS **110-a** based on checking the snapshots **235** for the file index corresponding to the file that includes the malware. In some examples, the alert **260** may identify the first snapshot that does not include the file as a safe recovery point. For example, if the file was introduced in the snapshot **235-b**, the safe recovery snapshot for the computing system **105-a** may be the snapshot **235-a**. In some examples, the DMS **110-a** may receive a command **265** from the computing device **115-a** to perform a recovery procedure for the computing system in response to the alert **260**. The command **265** may indicate the snapshot (e.g., the snapshot **235-a**) to use as the recovery snapshot. The DMS **110-a** may perform a recovery procedure for the computing system **105-a** using the indicated recovery snapshot in the command **265**.

[0050] In some cases, a full scan job of the DMS **110-a** may scan the most recent snapshot against new malware detection rules in the new malware intelligence information **215** received from the malware intelligence source **210**. In some examples, the full scan job may run periodically. In some examples, the full scan job may be triggered by reception of new malware intelligence information **215**. For example, either periodically or in response to the reception of the new malware intelligence information **215**, the full scan job may compare the new malware intelligence information **215** against the malware against the malware intelligence stored in the threat feed **220** which was used for a most recent full scan. If the new malware intelligence information **215** includes malware detection rules that were not included in the malware intelligence which was used for a most recent full scan, the full scan job of the DMS **110-a** may trigger a full scan for the snapshots associated with the customer (e.g., the snapshots **235** and the snapshots **240**) using the new malware intelligence information **215** (e.g.,

using the malware detection rules in the new malware intelligence information **215** that was not included in the malware intelligence which was used for a most recent full scan).

[0051] When a full scan is triggered, the full scan job may scan all of the files in the latest snapshot of each computing system associated with a customer against the new malware detection rules in the new malware intelligence information **215**.

[0052] A full scan may involve a full YARA scan, which may be performed by the YARA scan job **245**. For example, the new YARA threat intelligence in the new malware intelligence information **215** may be used to scan the full content of all of the files in the latest snapshot of each computing system **105** associated with the customer. For example, the latest snapshot of the computing system **105-a** may be the snapshot **235-n**, and the latest snapshot of the computing system **105-b** may be the snapshot **240-n**.

[0053] A full scan may involve checking, by the hash scan job **250** of the DMS **110-a** for each new indicator of malware in the new malware intelligence information **215** in the hash catalog **270** of the customer. The hash catalog **270** for a given customer may include a hash of each file included in the snapshots (e.g., the snapshots **235** and the snapshots **240**) associated with the customer that were new or modified with respect to a prior snapshot (e.g., and all of the files in a base snapshot). When hashes are computed as part of an incremental scan in response to capturing a snapshot, the hashes may be added to the hash catalog. Thus, the hash catalog may include hashes of each modified or added file in each snapshot for a customer. In some examples, the hash catalog may be maintained at a cloud storage environment. The hashes indicative of malware in the new malware intelligence information **215** may be compared to the hash catalog for a customer to determine if the hashes of any files in the hash catalog match known indicators of malware. A match may indicate that the corresponding file includes malware.

[0054] If malware is detected in a file based on a full scan job, an alert **260** may be provided to a computing device **115-a** associated with an administrative account of the customer. For example, the alert **260** may be an email or a message on a user interface at the computing device **115-a** for interfacing with the DMS **110-a** for the customer. The alert **260** may indicate that the file includes malware, and may indicate a first snapshot of the corresponding computing system **105** that included the file. Accordingly, the customer (e.g., the administrator) may identify a first snapshot that does not include the file as a safe recovery point for the corresponding computing system. In some examples, the first snapshot of the corresponding computing system **105** that included the file may be identified by the DMS **110-a** based on checking the snapshots **235** for the file index corresponding to the file that includes the malware. In some examples, the alert **260** may identify the first snapshot that does not include the file as a safe recovery point. For example, if the file was introduced in snapshot **235-b**, the safe recovery snapshot for the computing system **105-a** may be the snapshot **235-a**. In some examples, the DMS **110-a** may receive a command **265** from the computing device **115-a** to perform a recovery procedure for the computing system in response to the alert **260**. The command **265** may indicate the snapshot (e.g., the snapshot **235-a**) to use as the recovery snapshot. The DMS **110-a** may perform a recovery

procedure for the computing system **105-a** using the indicated recovery snapshot in the command **265**.

[0055] The combination of the incremental scan job being performed on each new snapshot, and the full scan for new malware intelligence being performed on the most recent snapshots of the computing systems **105** (for YARA rules) and the hash catalog **270** (for hashes) may ensure that all files associated with a customer that are backed up by the DMS **110-a** may be scanned for the latest malware intelligence received by the DMS **110-a**.

[0056] FIG. 3 shows an example of a computing environment **300** that supports malware monitoring and detection in accordance with aspects of the present disclosure. The computing environment **300** may implement one or more aspects of the computing environment **100** or the computing environment **200**. For example, the computing environment **300** includes a DMS **110-b** which may be an example of the DMS **110** as described herein.

[0057] The DMS **110-b** may provide backup services for multiple customers (e.g., a customer A and a customer B). The DMS **110-b** may capture and store the snapshots **335** of a first computing system of the customer A and the snapshots **340** of a second computing system of the customer A. The DMS **110-b** may store the snapshots **335** and the snapshots **340** in a first storage environment **305-a** for the customer A. The DMS **110-b** may capture and store the snapshots **345** of a first computing system of the customer B and the snapshots **350** of a second computing system of the customer B. The DMS **110-b** may store the snapshots **345** and the snapshots **340** in a second storage environment **305-b** for the customer B. In some examples, the first storage environment **305-a** and the second storage environment **305-b** may be one or more storage nodes **185** at the DMS **110-a** as described with reference to FIG. 1. In some examples, the first storage environment **305-a** and the second storage environment **305-b** be an external storage environments accessible to the DMS **110-a** such as the cloud environment **195** as described with reference to FIG. 1.

[0058] The DMS **110-b** may receive malware intelligence information from multiple malware intelligence sources. For example, a first malware intelligence source **210-a** may be associated with the customer A, a second malware intelligence source **210-b** may be associated with the customer B, and a third malware intelligence source **210-c** may be applicable or applied by the DMS **110-b** to multiple customers (e.g., all customers) of the DMS **110-b**.

[0059] The DMS **110-b** may maintain separate threat feeds for different customers. For example, the DMS **110-b** may maintain a first threat feed **220-a** for the customer A that may include malware intelligence information received from the first malware intelligence source **210-a** and/or the third malware intelligence source **210-c**. The DMS **110-b** may maintain a second threat feed **220-b** for the customer B that may include malware intelligence information received from the second malware intelligence source **210-b** and/or the third malware intelligence source **210-c**. Similarly, the DMS **110-b** may maintain a first hash catalog **270-a** for the customer A and a second hash catalog **270-b** for the customer B. The scan manager **225-a** of the DMS **110-b** may use the first threat feed **220-a** to run incremental scans on new snapshots of the snapshots **335** or the snapshots **340** or full scans on the snapshots **335** and the snapshots **340**. The scan manager **225-a** of the DMS **110-b** may use the first hash catalog **270-a** to perform a hash analysis for a full scan based

on reception of new malware intelligence information at the first threat feed **220-a** for the customer A. Similarly, the scan manager **225-a** of the DMS **110-b** may use the second threat feed **220-b** to run incremental scans on new snapshots of the snapshots **345** or the snapshots **340** or full scans on the snapshots **345** and the snapshots **350** for the customer B. The scan manager **225-a** of the DMS **110-b** may use the second hash catalog **270-b** to perform a hash analysis for a full scan based on reception of new malware intelligence information at the second threat feed **220-b** for the customer B. If the DMS **110-b** detects malware in a file of one of the customers, the DMS **110-b** may provide an alert to a computing device **115** associated with the corresponding customer as described herein.

[0060] FIG. 4 shows an example of a process flow **400** that supports malware monitoring and detection in accordance with aspects of the present disclosure. The process flow **400** may implement one or more aspects of the computing environment **100**, the computing environment **200**, or the computing environment **300**. For example, the process flow **400** may include a DMS **110-c**, a malware intelligence source **210-d**, and a computing device **115-b**, which may be examples of a DMS **110**, a malware intelligence source **210**, and a computing device **115** as described herein. In the following description of the process flow **400**, operations between the DMS **110-c**, the malware intelligence source **210-d**, and the computing device **115-b** may be added, omitted, or performed in a different order (with respect to the exemplary order shown).

[0061] At **405**, the DMS **110-c** may receive an indication of a set of malware detection rules associated with a customer of the DMS **110-c**. In some examples, the set of malware detection rules may include a first subset of malware detection rules specific to the customer and a second subset of malware detection rules common to a set of multiple customers of the DMS **110-c**. For example, the DMS **110-c** may receive the first subset from a malware intelligence subscription specific to the customer, and the DMS **110-c** may receive the second subset from malware intelligence subscription applicable to multiple customers of the DMS **110-c**.

[0062] At **410**, the DMS **110-c** may capture a first snapshot of a computing system associated with the customer.

[0063] At **415**, the DMS **110-c** may perform a check, based on capturing the snapshot, of a set of files within the first snapshot based on the set of malware detection rules. The set of files may include files that have been modified or added with respect to a second snapshot of the computing system prior in time to the first snapshot. In some examples, an incremental scan job may periodically check for new snapshots at the DMS **110-c**, and may trigger the check based on detecting that the DMS **110-c** captured the first snapshot.

[0064] In some examples, performing the check at **415** may include scanning the set of files against a set of YARA rules, the set of malware detection rules including the set of YARA rules. In some examples, performing the check at **415** may include: computing a respective set of hashes for the set of files; and performing a comparison of the respective set of hashes to a bloom filter representative of a set of hashes indicative of malware, the set of malware detection rules including the set of hashes indicative of malware.

[0065] At 420, the DMS 110-c may detect, based on the check at 415, that a file of the set of files includes first malware.

[0066] At 425, the DMS 110-c may identify, based on detecting at 420 that the file includes the first malware, a temporally first snapshot of the computing system that includes the file including the first malware.

[0067] In some examples, at 430, the DMS 110-c may transmit, to the computing device 115-b, an alert message that includes an indication that the file includes the first malware and an indication of the temporally first snapshot. The computing device 115-b may be associated with an administrative account associated with the customer.

[0068] In some examples, at 435, the DMS 110-c may receive, from the computing device 115-b and based on the alert message at 430, a recovery request message. The recovery request message may include an indication to perform a recovery procedure for the computing system using a recovery snapshot that is temporally prior to the temporally first snapshot.

[0069] In some examples, at 440, the DMS 110-c may perform the recovery procedure based on the recovery request message.

[0070] In some examples, the DMS 110-c may identify, based on detecting that the file includes the first malware, the recovery snapshot. The recovery snapshot may be the snapshot that is immediately prior in time to the temporally first snapshot in a chain of snapshots of the computing system. In some examples, the alert message at 430 may indicate the recovery snapshot identified by the DMS 110-c.

[0071] In some examples, the DMS 110-c may receive an indication of a second set of malware detection rules associated with a second customer of the DMS 110-c. The second set of malware detection rules may include a third subset of malware detection rules specific to the customer and the second subset of malware detection rules common to the multiple customers of the DMS 110-c. In some examples, the DMS 110-c may capture a third snapshot of a second computing system associated with the second customer. In such examples, the DMS 110-c may perform a second check, based on capturing the third snapshot, of a second set of files within the third snapshot based on the set of malware detection rules, where the second set of files may include files that have been modified or added with respect to a fourth snapshot of the second computing system prior in time to the third snapshot. In such examples, the DMS 110-c may detect, based on the check, that a second file of the set of files includes second malware. In such examples, the DMS 110-c may identify, based on detecting that the second file includes the second malware, a second temporally first snapshot of the second computing system that includes the second file including the second malware. In some examples, the DMS 110-c may transmit an alert to a second computing device associated with the second customer that includes an indication that the second file includes the second malware and an indication of the second temporally first snapshot.

[0072] FIG. 5 shows an example of a process flow 500 that supports malware monitoring and detection in accordance with aspects of the present disclosure. The process flow 500 may implement one or more aspects of the computing environment 100, the computing environment 200, or the computing environment 300. For example, the process flow 500 may include a DMS 110-d, a malware intelligence

source 210-e, and a computing device 115-c, which may be examples of a DMS 110, a malware intelligence source 210, and a computing device 115 as described herein. In the following description of the process flow 500, operations between the DMS 110-d, the malware intelligence source 210-e, and the computing device 115-c may be added, omitted, or performed in a different order (with respect to the exemplary order shown).

[0073] At 505, the DMS 110-d may receive an indication of a set of malware detection rules associated with a customer of the DMS 110-d. In some examples, the set of malware detection rules may include a first subset of malware detection rules specific to the customer and a second subset of malware detection rules common to a set of multiple customers of the DMS 110-d. For example, the DMS 110-d may receive the first subset from a malware intelligence subscription specific to the customer, and the DMS 110-d may receive the second subset from malware intelligence subscription applicable to multiple customers of the DMS 110-c.

[0074] At 510, the DMS 110-d may perform a check, based on reception of the set of malware detection rules, of respective sets of files within a respective set of most recent snapshots of a set of computing systems associated with the customer based on the set of malware detection rules.

[0075] At 515 the DMS 110-d may detect, based on the check at 510, that a file of a respective set of files within a first most recent snapshot of the respective set of most recent snapshots corresponding to a first respective computing system of the set of computing systems includes first malware.

[0076] At 520, the DMS 110-d may identify, based on detecting that the file includes the first malware, a temporally first snapshot of the first respective computing system that includes the file including the first malware.

[0077] In some examples, at 525, the DMS 110-d may transmit, to the computing device 115-c, an alert message that includes an indication that the file includes the first malware and an indication of the temporally first snapshot. The computing device 115-c may be associated with an administrative account associated with the customer.

[0078] In some examples, at 530, the DMS 110-d may receive, from the computing device 115-b and based on the alert message at 525, a recovery request message. The recovery request message may include an indication to perform a recovery procedure for the first respective computing system using a recovery snapshot that is temporally prior to the temporally first snapshot.

[0079] In some examples, at 535, the DMS 110-d may perform the recovery procedure based on the recovery request message.

[0080] In some examples, the DMS 110-d may identify, based on detecting that the file includes the first malware, the recovery snapshot. The recovery snapshot may be the snapshot that is immediately prior in time to the temporally first snapshot in a chain of snapshots of the first respective computing system. In some examples, the alert message at 430 may indicate the recovery snapshot identified by the DMS 110-d.

[0081] In some examples, the DMS 110-d may identify that a time period since a last check of the respective sets of files within the respective set of most recent snapshots of the set of computing systems satisfies a threshold time period,

and performance of the check at **510** may be based on identifying that the time period satisfies the threshold time period,

[0082] In some examples, performing the check at **510** may include scanning the respective sets of files against a set of YARA rules, the set of malware detection rules including the set of YARA rules. In some examples, the set of YARA rules may be a subset of YARA rules received in the malware detection rules that are different from a prior set of YARA rules stored at a threat feed at the DMS **110-d**.

[0083] In some examples, performing the check at **510** may include performing a comparison of a hash catalog associated with the customer to a set of hashes indicative of malware, the set of malware detection rules including the set of hashes indicative of malware. For example, the hash catalog associated with the customer may include hashes of each modified or added file in each snapshot of the set of computing systems.

[0084] FIG. 6 shows a block diagram **600** of a system **605** that supports malware monitoring and detection in accordance with aspects of the present disclosure. In some examples, the system **605** may be an example of aspects of one or more components described with reference to FIG. 1, such as a DMS **110**. The system **605** may include an input interface **610**, an output interface **615**, and a DMS manager **620**. The system **605** may also include one or more processors. Each of these components may be in communication with one another (e.g., via one or more buses, communications links, communications interfaces, or any combination thereof).

[0085] The input interface **610** may manage input signaling for the system **605**. For example, the input interface **610** may receive input signaling (e.g., messages, packets, data, instructions, commands, or any other form of encoded information) from other systems or devices. The input interface **610** may send signaling corresponding to (e.g., representative of or otherwise based on) such input signaling to other components of the system **605** for processing. For example, the input interface **610** may transmit such corresponding signaling to the DMS manager **620** to support malware monitoring and detection. In some cases, the input interface **610** may be a component of a network interface **825** as described with reference to FIG. 8.

[0086] The output interface **615** may manage output signaling for the system **605**. For example, the output interface **615** may receive signaling from other components of the system **605**, such as the DMS manager **620**, and may transmit such output signaling corresponding to (e.g., representative of or otherwise based on) such signaling to other systems or devices. In some cases, the output interface **615** may be a component of a network interface **825** as described with reference to FIG. 8.

[0087] For example, the DMS manager **620** may include a malware detection rule manager **625**, a snapshot acquisition manager **630**, a malware scan manager **635**, a malware detection manager **640**, a malware file identification manager **645**, or any combination thereof. In some examples, the DMS manager **620**, or various components thereof, may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the input interface **610**, the output interface **615**, or both. For example, the DMS manager **620** may receive information from the input interface **610**, send information to the output interface **615**, or be integrated in combination

with the input interface **610**, the output interface **615**, or both to receive information, transmit information, or perform various other operations as described herein.

[0088] The malware detection rule manager **625** may be configured as or otherwise support a means for receiving, by a DMS, an indication of a set of malware detection rules associated with a customer of the DMS. The snapshot acquisition manager **630** may be configured as or otherwise support a means for capturing, by the DMS, a first snapshot of a computing system associated with the customer. The malware scan manager **635** may be configured as or otherwise support a means for performing a check, by the DMS and based on capturing the first snapshot, of a set of files within the first snapshot based on the set of malware detection rules, where the set of files include files that have been modified or added with respect to a second snapshot of the computing system prior in time to the first snapshot. The malware detection manager **640** may be configured as or otherwise support a means for detecting, by the DMS and based on the check, that a file of the set of files includes first malware. The malware file identification manager **645** may be configured as or otherwise support a means for identifying, by the DMS based on detecting that the file includes the first malware, a temporally first snapshot of the computing system that includes the file including the first malware.

[0089] FIG. 7 shows a block diagram **700** of a DMS manager **720** that supports malware monitoring and detection in accordance with aspects of the present disclosure. The DMS manager **720** may be an example of aspects of a DMS or a DMS manager **620**, or both, as described herein. The DMS manager **720**, or various components thereof, may be an example of means for performing various aspects of malware monitoring and detection as described herein. For example, the DMS manager **720** may include a malware detection rule manager **725**, a snapshot acquisition manager **730**, a malware scan manager **735**, a malware detection manager **740**, a malware file identification manager **745**, a malware alert manager **750**, a YARA rule manager **755**, a file hash manager **760**, a hash catalog manager **765**, a recovery request manager **770**, a recovery manager **775**, a periodic malware scan manager **780**, a malware hash manager **785**, a recovery snapshot identification manager **790**, or any combination thereof. Each of these components, or components of subcomponents thereof (e.g., one or more processors, one or more memories), may communicate, directly or indirectly, with one another (e.g., via one or more buses, communications links, communications interfaces, or any combination thereof).

[0090] The malware detection rule manager **725** may be configured as or otherwise support a means for receiving, by a DMS, an indication of a set of malware detection rules associated with a customer of the DMS. The snapshot acquisition manager **730** may be configured as or otherwise support a means for capturing, by the DMS, a first snapshot of a computing system associated with the customer. The malware scan manager **735** may be configured as or otherwise support a means for performing a check, by the DMS and based on capturing the first snapshot, of a set of files within the first snapshot based on the set of malware detection rules, where the set of files include files that have been modified or added with respect to a second snapshot of the computing system prior in time to the first snapshot. The malware detection manager **740** may be configured as or otherwise support a means for detecting, by the DMS and

based on the check, that a file of the set of files includes first malware. The malware file identification manager **745** may be configured as or otherwise support a means for identifying, by the DMS based on detecting that the file includes the first malware, a temporally first snapshot of the computing system that includes the file including the first malware.

[0091] In some examples, the malware alert manager **750** may be configured as or otherwise support a means for transmitting, by the DMS to a computing device associated with an administrative account associated with the customer, an alert message including an indication that the file includes the first malware and an indication of the temporally first snapshot.

[0092] In some examples, the recovery request manager **770** may be configured as or otherwise support a means for receiving, by the DMS from the computing device and based on the alert message, a recovery request message, where the recovery request message includes an indication to perform a recovery procedure for the computing system using a recovery snapshot that is temporally prior to the temporally first snapshot. In some examples, the recovery manager **775** may be configured as or otherwise support a means for performing, by the DMS, the recovery procedure based on the recovery request message.

[0093] In some examples, the recovery snapshot identification manager **790** may be configured as or otherwise support a means for identifying, by the DMS and based on detecting that the file includes the first malware, the recovery snapshot, where the recovery snapshot is immediately prior in time to the temporally first snapshot in a chain of snapshots of the computing system, where the alert message further includes an indication of the recovery snapshot.

[0094] In some examples, the malware scan manager **735** may be configured as or otherwise support a means for performing a second check, by the DMS and based on reception of the set of malware detection rules, of respective sets of files within a respective set of most recent snapshots of a set of computing systems associated with the customer based on the set of malware detection rules, the set of computing systems including the computing system.

[0095] In some examples, the malware detection rule manager **725** may be configured as or otherwise support a means for detecting, by the DMS and based on the second check, that a second file of a respective set of files within a first most recent snapshot of the respective set of most recent snapshots corresponding to a first respective computing system of the set of computing systems includes second malware. In some examples, the malware file identification manager **745** may be configured as or otherwise support a means for identifying, by the DMS and based on detecting that the second file includes the second malware, a second temporally first snapshot of the first respective computing system that includes the second file including the second malware.

[0096] In some examples, the malware alert manager **750** may be configured as or otherwise support a means for transmitting, by the DMS to a computing device associated with an administrative account associated with the customer, an alert message including an indication that the second file includes the second malware and an indication of the second temporally first snapshot.

[0097] In some examples, the periodic malware scan manager **780** may be configured as or otherwise support a means for identifying, by the DMS, that a time period since a last

check of the respective sets of files within the respective set of most recent snapshots of the set of computing systems satisfies a threshold time period, where performance of the second check is based on identifying that the time period satisfies the threshold time period.

[0098] In some examples, to support performing the second check, the YARA rule manager **755** may be configured as or otherwise support a means for scanning the respective sets of files against a set of YARA rules, the set of malware detection rules including the set of YARA rules.

[0099] In some examples, to support performing the second check, the hash catalog manager **765** may be configured as or otherwise support a means for performing a comparison of a hash catalog associated with the customer to a set of hashes indicative of malware, the set of malware detection rules including the set of hashes indicative of malware.

[0100] In some examples, to support performing the check, the YARA rule manager **755** may be configured as or otherwise support a means for scanning the set of files against a set of YARA rules, the set of malware detection rules including the set of YARA rules.

[0101] In some examples, to support performing the second check, the file hash manager **760** may be configured as or otherwise support a means for computing a respective set of hashes for the set of files. In some examples, to support performing the check, the malware hash manager **785** may be configured as or otherwise support a means for performing a comparison of the respective set of hashes to a bloom filter representative of a set of hashes indicative of malware, the set of malware detection rules including the set of hashes indicative of malware.

[0102] In some examples, the malware detection manager **740** may be configured as or otherwise support a means for identifying, based on the comparison, that a subset of hashes of the respective set of hashes match known malware. In some examples, the malware detection manager **740** may be configured as or otherwise support a means for checking a subset of the set of files corresponding to the subset of hashes against a set of known malware.

[0103] In some examples, the set of malware detection rules include a first subset of malware detection rules specific to the customer and a second subset of malware detection rules common to a set of multiple customers of the DMS.

[0104] In some examples, the malware detection rule manager **725** may be configured as or otherwise support a means for receiving, by the DMS, an indication of a second set of malware detection rules associated with a second customer of the DMS, where the second set of malware detection rules include a third subset of malware detection rules specific to the customer and the second subset of malware detection rules.

[0105] In some examples, the snapshot acquisition manager **730** may be configured as or otherwise support a means for capturing, by the DMS, a third snapshot of a second computing system associated with the second customer. In some examples, the malware scan manager **735** may be configured as or otherwise support a means for performing a second check, by the DMS and based on capturing the third snapshot, of a second set of files within the third snapshot based on the set of malware detection rules, where the second set of files include files that have been modified or added with respect to a fourth snapshot of the second computing system prior in time to the third snapshot. In

some examples, the malware detection manager **740** may be configured as or otherwise support a means for detecting, by the DMS and based on the check, that a second file of the set of files includes second malware. In some examples, the malware file identification manager **745** may be configured as or otherwise support a means for identifying, by the DMS based on detecting that the second file includes the second malware, a second temporally first snapshot of the second computing system that includes the second file including the second malware.

[0106] FIG. 8 shows a block diagram **800** of a system **805** that supports malware monitoring and detection in accordance with aspects of the present disclosure. The system **805** may be an example of or include components of a system **605** as described herein. The system **805** may include components for data management, including components such as a DMS manager **820**, an input information **810**, an output information **815**, a network interface **825**, at least one memory **830**, at least one processor **835**, and a storage **840**. These components may be in electronic communication or otherwise coupled with each other (e.g., operatively, communicatively, functionally, electronically, electrically; via one or more buses, communications links, communications interfaces, or any combination thereof). Additionally, the components of the system **805** may include corresponding physical components or may be implemented as corresponding virtual components (e.g., components of one or more virtual machines). In some examples, the system **805** may be an example of aspects of one or more components described with reference to FIG. 1, such as a DMS **110**.

[0107] The network interface **825** may enable the system **805** to exchange information (e.g., input information **810**, output information **815**, or both) with other systems or devices (not shown). For example, the network interface **825** may enable the system **805** to connect to a network (e.g., a network **120** as described herein). The network interface **825** may include one or more wireless network interfaces, one or more wired network interfaces, or any combination thereof. In some examples, the network interface **825** may be an example of may be an example of aspects of one or more components described with reference to FIG. 1, such as one or more network interfaces **165**.

[0108] Memory **830** may include RAM, ROM, or both. The memory **830** may store computer-readable, computer-executable software including instructions that, when executed, cause the processor **835** to perform various functions described herein. In some cases, the memory **830** may contain, among other things, a basic input/output system (BIOS), which may control basic hardware or software operation such as the interaction with peripheral components or devices. In some cases, the memory **830** may be an example of aspects of one or more components described with reference to FIG. 1, such as one or more memories **175**.

[0109] The processor **835** may include an intelligent hardware device, (e.g., a general-purpose processor, a DSP, a CPU, a microcontroller, an ASIC, a field programmable gate array (FPGA), a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). The processor **835** may be configured to execute computer-readable instructions stored in a memory **830** to perform various functions (e.g., functions or tasks supporting malware monitoring and detection). Though a single processor **835** is depicted in the example of FIG. 8, it is to be understood that the system **805**

may include any quantity of one or more of processors **835** and that a group of processors **835** may collectively perform one or more functions ascribed herein to a processor, such as the processor **835**. In some cases, the processor **835** may be an example of aspects of one or more components described with reference to FIG. 1, such as one or more processors **170**.

[0110] Storage **840** may be configured to store data that is generated, processed, stored, or otherwise used by the system **805**. In some cases, the storage **840** may include one or more HDDs, one or more SSDs, or both. In some examples, the storage **840** may be an example of a single database, a distributed database, multiple distributed databases, a data store, a data lake, or an emergency backup database. In some examples, the storage **840** may be an example of one or more components described with reference to FIG. 1, such as one or more network disks **180**.

[0111] For example, the DMS manager **820** may be configured as or otherwise support a means for receiving, by a DMS, an indication of a set of malware detection rules associated with a customer of the DMS. The DMS manager **820** may be configured as or otherwise support a means for capturing, by the DMS, a first snapshot of a computing system associating with the customer. The DMS manager **820** may be configured as or otherwise support a means for performing a check, by the DMS and based on capturing the first snapshot, of a set of files within the first snapshot based on the set of malware detection rules, where the set of files include files that have been modified or added with respect to a second snapshot of the computing system prior in time to the first snapshot. The DMS manager **820** may be configured as or otherwise support a means for detecting, by the DMS and based on the check, that a file of the set of files includes first malware. DMS manager **820** may be configured as or otherwise support a means for identifying, by the DMS based on detecting that the file includes the first malware, a temporally first snapshot of the computing system that includes the file including the first malware.

[0112] By including or configuring the DMS manager **820** in accordance with examples as described herein, the system **805** may support techniques for malware monitoring and detection, which may provide one or more benefits such as, for example, improved reliability, improved user experience, more efficient utilization of computing resources, network resources or both, improved scalability, or improved security, among other possibilities.

[0113] FIG. 9 shows a flowchart illustrating a method **900** that supports malware monitoring and detection in accordance with aspects of the present disclosure. The operations of the method **900** may be implemented by a DMS or its components as described herein. For example, the operations of the method **900** may be performed by a DMS as described with reference to FIGS. 1 through 8. In some examples, a DMS may execute a set of instructions to control the functional elements of the DMS to perform the described functions. Additionally, or alternatively, the DMS may perform aspects of the described functions using special-purpose hardware.

[0114] At **905**, the method may include receiving, by a DMS, an indication of a set of malware detection rules associated with a customer of the DMS. The operations of **905** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the opera-

tions of **905** may be performed by a malware detection rule manager **725** as described with reference to FIG. 7.

[0115] At **910**, the method may include capturing, by the DMS, a first snapshot of a computing system associated with the customer. The operations of **910** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **910** may be performed by a snapshot acquisition manager **730** as described with reference to FIG. 7.

[0116] At **915**, the method may include performing a check, by the DMS and based on capturing the first snapshot, of a set of files within the first snapshot based on the set of malware detection rules, where the set of files include files that have been modified or added with respect to a second snapshot of the computing system prior in time to the first snapshot. The operations of **915** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **915** may be performed by a malware scan manager **735** as described with reference to FIG. 7.

[0117] At **920**, the method may include detecting, by the DMS and based on the check, that a file of the set of files includes first malware. The operations of **920** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **920** may be performed by a malware detection manager **740** as described with reference to FIG. 7.

[0118] At **925**, the method may include identifying, by the DMS based on detecting that the file includes the first malware, a temporally first snapshot of the computing system that includes the file including the first malware. The operations of **925** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **925** may be performed by a malware file identification manager **745** as described with reference to FIG. 7.

[0119] FIG. 10 shows a flowchart illustrating a method **1000** that supports malware monitoring and detection in accordance with aspects of the present disclosure. The operations of the method **1000** may be implemented by a DMS or its components as described herein. For example, the operations of the method **1000** may be performed by a DMS as described with reference to FIGS. 1 through 8. In some examples, a DMS may execute a set of instructions to control the functional elements of the DMS to perform the described functions. Additionally, or alternatively, the DMS may perform aspects of the described functions using special-purpose hardware.

[0120] At **1005**, the method may include receiving, by a DMS, an indication of a set of malware detection rules associated with a customer of the DMS. The operations of **1005** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1005** may be performed by a malware detection rule manager **725** as described with reference to FIG. 7.

[0121] At **1010**, the method may include capturing, by the DMS, a first snapshot of a computing system associated with the customer. The operations of **1010** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1010** may be performed by a snapshot acquisition manager **730** as described with reference to FIG. 7.

[0122] At **1015**, the method may include performing a check, by the DMS and based on capturing the first snapshot,

of a set of files within the first snapshot based on the set of malware detection rules, where the set of files include files that have been modified or added with respect to a second snapshot of the computing system prior in time to the first snapshot. The operations of **1015** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1015** may be performed by a malware scan manager **735** as described with reference to FIG. 7.

[0123] At **1020**, the method may include detecting, by the DMS and based on the check, that a file of the set of files includes first malware. The operations of **1020** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1020** may be performed by a malware detection manager **740** as described with reference to FIG. 7.

[0124] At **1025**, the method may include identifying, by the DMS based on detecting that the file includes the first malware, a temporally first snapshot of the computing system that includes the file including the first malware. The operations of **1025** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1025** may be performed by a malware file identification manager **745** as described with reference to FIG. 7.

[0125] At **1030**, the method may include transmitting, by the DMS to a computing device associated with an administrative account associated with the customer, an alert message including an indication that the file includes the first malware and an indication of the temporally first snapshot. The operations of **1030** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1030** may be performed by a malware alert manager **750** as described with reference to FIG. 7.

[0126] FIG. 11 shows a flowchart illustrating a method **1100** that supports malware monitoring and detection in accordance with aspects of the present disclosure. The operations of the method **1100** may be implemented by a DMS or its components as described herein. For example, the operations of the method **1100** may be performed by a DMS as described with reference to FIGS. 1 through 8. In some examples, a DMS may execute a set of instructions to control the functional elements of the DMS to perform the described functions. Additionally, or alternatively, the DMS may perform aspects of the described functions using special-purpose hardware.

[0127] At **1105**, the method may include receiving, by a DMS, an indication of a set of malware detection rules associated with a customer of the DMS. The operations of **1105** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1105** may be performed by a malware detection rule manager **725** as described with reference to FIG. 7.

[0128] At **1110**, the method may include capturing, by the DMS, a first snapshot of a computing system associated with the customer. The operations of **1110** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1110** may be performed by a snapshot acquisition manager **730** as described with reference to FIG. 7.

[0129] At **1115**, the method may include performing a check, by the DMS and based on capturing the first snapshot, of a set of files within the first snapshot based on the set of

malware detection rules, where the set of files include files that have been modified or added with respect to a second snapshot of the computing system prior in time to the first snapshot. The operations of **1115** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1115** may be performed by a malware scan manager **735** as described with reference to FIG. 7.

[0130] At **1120**, the method may include detecting, by the DMS and based on the check, that a file of the set of files includes first malware. The operations of **1120** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1120** may be performed by a malware detection manager **740** as described with reference to FIG. 7.

[0131] At **1125**, the method may include identifying, by the DMS and based on the check, that the file includes the first malware, a temporally first snapshot of the computing system that includes the file including the first malware. The operations of **1125** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1125** may be performed by a malware file identification manager **745** as described with reference to FIG. 7.

[0132] At **1130**, the method may include performing a second check, by the DMS and based on reception of the set of malware detection rules, of respective sets of files within a respective set of most recent snapshots of a set of computing systems associated with the customer based on the set of malware detection rules, the set of computing systems including the computing system. The operations of **1130** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1130** may be performed by a malware scan manager **735** as described with reference to FIG. 7.

[0133] A method by an apparatus is described. The method may include receiving, by a DMS, an indication of a set of malware detection rules associated with a customer of the DMS, capturing, by the DMS, a first snapshot of a computing system associated with the customer, performing a check, by the DMS and based on capturing the first snapshot, of a set of files within the first snapshot based on the set of malware detection rules, where the set of files include files that have been modified or added with respect to a second snapshot of the computing system prior in time to the first snapshot, detecting, by the DMS and based on the check, that a file of the set of files includes first malware, and identifying, by the DMS based on detecting that the file includes the first malware, a temporally first snapshot of the computing system that includes the file including the first malware.

[0134] An apparatus is described. The apparatus may include one or more memories storing processor executable code, and one or more processors coupled with the one or more memories. The one or more processors may individually or collectively be operable to execute the code to cause the apparatus to receive, by a DMS, an indication of a set of malware detection rules associated with a customer of the DMS, capturing, by the DMS, a first snapshot of a computing system associate with the customer, perform a check, by the DMS and based on capturing the first snapshot, of a set of files within the first snapshot based on the set of malware detection rules, where the set of files include files that have been modified or added with respect to a second snapshot of

the computing system prior in time to the first snapshot, detect, by the DMS and based on the check, that a file of the set of files includes first malware, and identify, by the DMS based on detecting that the file includes the first malware, a temporally first snapshot of the computing system that includes the file including the first malware.

[0135] Another apparatus is described. The apparatus may include means for receiving, by a DMS, an indication of a set of malware detection rules associated with a customer of the DMS, means for capturing, by the DMS, a first snapshot of a computing system associated with the customer, means for performing a check, by the DMS and based on capturing the first snapshot, of a set of files within the first snapshot based on the set of malware detection rules, where the set of files include files that have been modified or added with respect to a second snapshot of the computing system prior in time to the first snapshot, means for detecting, by the DMS and based on the check, that a file of the set of files includes first malware, and means for identifying, by the DMS based on detecting that the file includes the first malware, a temporally first snapshot of the computing system that includes the file including the first malware.

[0136] A non-transitory computer-readable medium storing code is described. The code may include instructions executable by one or more processors to receive, by a DMS, an indication of a set of malware detection rules associated with a customer of the DMS, capturing, by the DMS, a first snapshot of a computing system associate with the customer, perform a check, by the DMS and based on capturing the first snapshot, of a set of files within the first snapshot based on the set of malware detection rules, where the set of files include files that have been modified or added with respect to a second snapshot of the computing system prior in time to the first snapshot, detect, by the DMS and based on the check, that a file of the set of files includes first malware, and identify, by the DMS based on detecting that the file includes the first malware, a temporally first snapshot of the computing system that includes the file including the first malware.

[0137] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting, by the DMS to a computing device associated with an administrative account associated with the customer, an alert message including an indication that the file includes the first malware and an indication of the temporally first snapshot.

[0138] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, by the DMS from the computing device and based on the alert message, a recovery request message, where the recovery request message includes an indication to perform a recovery procedure for the computing system using a recovery snapshot that may be temporally prior to the temporally first snapshot and performing, by the DMS, the recovery procedure based on the recovery request message.

[0139] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying, by the DMS and based on detecting that the file includes the first malware, the recovery snapshot, where the recovery snapshot may be immediately prior

in time to the temporally first snapshot in a chain of snapshots of the computing system, where the alert message further includes an indication of the recovery snapshot.

[0140] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for performing a second check, by the DMS and based on reception of the set of malware detection rules, of respective sets of files within a respective set of most recent snapshots of a set of computing systems associated with the customer based on the set of malware detection rules, the set of computing systems including the computing system.

[0141] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for detecting, by the DMS and based on the second check, that a second file of a respective set of files within a first most recent snapshot of the respective set of most recent snapshots corresponding to a first respective computing system of the set of computing systems includes second malware and identifying, by the DMS and based on detecting that the second file includes the second malware, a second temporally first snapshot of the first respective computing system that includes the second file including the second malware.

[0142] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting, by the DMS to a computing device associated with an administrative account associated with the customer, an alert message including an indication that the second file includes the second malware and an indication of the second temporally first snapshot.

[0143] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying, by the DMS, that a time period since a last check of the respective sets of files within the respective set of most recent snapshots of the set of computing systems satisfies a threshold time period, where performance of the second check may be based on identifying that the time period satisfies the threshold time period.

[0144] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, performing the second check may include operations, features, means, or instructions for scanning the respective sets of files against a set of YARA rules, the set of malware detection rules including the set of YARA rules.

[0145] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, performing the second check may include operations, features, means, or instructions for performing a comparison of a hash catalog associated with the customer to a set of hashes indicative of malware, the set of malware detection rules including the set of hashes indicative of malware.

[0146] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, performing the check may include operations, features, means, or instructions for scanning the set of files against a set of YARA rules, the set of malware detection rules including the set of YARA rules.

[0147] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, performing the check may include operations, features,

means, or instructions for co computing a respective set of hashes for the set of files and performing a comparison of the respective set of hashes to a bloom filter representative of a set of hashes indicative of malware, the set of malware detection rules including the set of hashes indicative of malware.

[0148] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying, based on the comparison, that a subset of hashes of the respective set of hashes match known malware and checking a subset of the set of files corresponding to the subset of hashes against a set of known malware.

[0149] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the set of malware detection rules include a first subset of malware detection rules specific to the customer and a second subset of malware detection rules common to a set of multiple customers of the DMS.

[0150] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, by the DMS, an indication of a second set of malware detection rules associated with a second customer of the DMS, where the second set of malware detection rules include a third subset of malware detection rules specific to the customer and the second subset of malware detection rules.

[0151] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, capturing, by the DMS, a third snapshot of a second computing system associated with the second customer, performing a second check, by the DMS and based on capturing the third snapshot, of a second set of files within the third snapshot based on the set of malware detection rules, where the second set of files include files that may have been modified or added with respect to a fourth snapshot of the second computing system prior in time to the third snapshot, detecting, by the DMS and based on the check, that a second file of the set of files includes second malware, and identifying, by the DMS based on detecting that the second file includes the second malware, a second temporally first snapshot of the second computing system that includes the second file including the second malware.

[0152] It should be noted that the methods described above describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Furthermore, aspects from two or more of the methods may be combined.

[0153] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

[0154] In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

[0155] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0156] The various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

[0157] The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described above can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations. Further, a system as used herein may be a collection of devices, a single device, or aspects within a single device.

[0158] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, non-transitory computer-readable media can comprise RAM, ROM, EEPROM) compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other

remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.

[0159] As used herein, including in the claims, the article “a” before a noun is open-ended and understood to refer to “at least one” of those nouns or “one or more” of those nouns. Thus, the terms “a,” “at least one,” “one or more,” and “at least one of one or more” may be interchangeable. For example, if a claim recites “a component” that performs one or more functions, each of the individual functions may be performed by a single component or by any combination of multiple components. Thus, “a component” having characteristics or performing functions may refer to “at least one of one or more components” having a particular characteristic or performing a particular function. Subsequent reference to a component introduced with the article “a” using the terms “the” or “said” refers to any or all of the one or more components. For example, a component introduced with the article “a” shall be understood to mean “one or more components,” and referring to “the component” subsequently in the claims shall be understood to be equivalent to referring to “at least one of the one or more components.”

[0160] Also, as used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

[0161] The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

What is claimed is:

1. A method, comprising:

receiving, by a data management system (DMS), an indication of a set of malware detection rules associated with a customer of the DMS;

capturing, by the DMS, a first snapshot of a computing system associated with the customer;

performing a check, by the DMS and based at least in part on capturing the first snapshot, of a set of files within the first snapshot based at least in part on the set of malware detection rules, wherein the set of files comprise files that have been modified or added with

respect to a second snapshot of the computing system prior in time to the first snapshot;

detecting, by the DMS and based at least in part on the check, that a file of the set of files comprises first malware; and

identifying, by the DMS based at least in part on detecting that the file comprises the first malware, a temporally first snapshot of the computing system that includes the file comprising the first malware.

2. The method of claim 1, further comprising: transmitting, by the DMS to a computing device associated with an administrative account associated with the customer, an alert message comprising an indication that the file comprises the first malware and an indication of the temporally first snapshot.

3. The method of claim 2, further comprising: receiving, by the DMS from the computing device and based at least in part on the alert message, a recovery request message, wherein the recovery request message comprises an indication to perform a recovery procedure for the computing system using a recovery snapshot that is temporally prior to the temporally first snapshot; and

performing, by the DMS, the recovery procedure based at least in part on the recovery request message.

4. The method of claim 3, further comprising: identifying, by the DMS and based at least in part on detecting that the file comprises the first malware, the recovery snapshot, wherein the recovery snapshot is immediately prior in time to the temporally first snapshot in a chain of snapshots of the computing system, wherein the alert message further comprises an indication of the recovery snapshot.

5. The method of claim 1, further comprising: performing a second check, by the DMS and based at least in part on reception of the set of malware detection rules, of respective sets of files within a respective set of most recent snapshots of a set of computing systems associated with the customer based at least in part on the set of malware detection rules, the set of computing systems comprising the computing system.

6. The method of claim 5, further comprising: detecting, by the DMS and based on the second check, that a second file of a respective set of files within a first most recent snapshot of the respective set of most recent snapshots corresponding to a first respective computing system of the set of computing systems comprises second malware; and

identifying, by the DMS and based on detecting that the second file comprises the second malware, a second temporally first snapshot of the first respective computing system that includes the second file comprising the second malware.

7. The method of claim 6, further comprising: transmitting, by the DMS to a computing device associated with an administrative account associated with the customer, an alert message comprising an indication that the second file comprises the second malware and an indication of the second temporally first snapshot.

8. The method of claim 5, further comprising: identifying, by the DMS, that a time period since a last check of the respective sets of files within the respective set of most recent snapshots of the set of computing systems satisfies a threshold time period, wherein per-

formance of the second check is based at least in part on identifying that the time period satisfies the threshold time period.

9. The method of claim 5, wherein performing the second check comprises:

scanning the respective sets of files against a set of YARA rules, the set of malware detection rules comprising the set of YARA rules.

10. The method of claim 5, wherein performing the second check comprises:

performing a comparison of a hash catalog associated with the customer to a set of hashes indicative of malware, the set of malware detection rules comprising the set of hashes indicative of malware.

11. The method of claim 1, wherein performing the check comprises:

scanning the set of files against a set of YARA rules, the set of malware detection rules comprising the set of YARA rules.

12. The method of claim 1, wherein performing the check comprises:

computing a respective set of hashes for the set of files; and

performing a comparison of the respective set of hashes to a bloom filter representative of a set of hashes indicative of malware, the set of malware detection rules comprising the set of hashes indicative of malware.

13. The method of claim 12, further comprising:

identifying, based at least in part on the comparison, that a subset of hashes of the respective set of hashes match known malware; and

checking a subset of the set of files corresponding to the subset of hashes against a set of known malware.

14. The method of claim 1, wherein the set of malware detection rules comprise a first subset of malware detection rules specific to the customer and a second subset of malware detection rules common to a plurality of customers of the DMS.

15. The method of claim 14, further comprising:

receiving, by the DMS, an indication of a second set of malware detection rules associated with a second customer of the DMS, wherein the second set of malware detection rules comprise a third subset of malware detection rules specific to the customer and the second subset of malware detection rules.

16. The method of claim 15, further comprising:

capturing, by the DMS, a third snapshot of a second computing system associated with the second customer;

performing a second check, by the DMS and based at least in part on capturing the third snapshot, of a second set of files within the third snapshot based at least in part on the set of malware detection rules, wherein the second set of files comprise files that have been modified or added with respect to a fourth snapshot of the second computing system prior in time to the third snapshot;

detecting, by the DMS and based at least in part on the check, that a second file of the set of files comprises second malware; and

identifying, by the DMS based at least in part on detecting that the second file comprises the second malware, a

second temporally first snapshot of the second computing system that includes the second file comprising the second malware.

17. An apparatus, comprising:

one or more memories storing processor-executable code; and

one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the apparatus to:

receive, by a data management system (DMS), an indication of a set of malware detection rules associated with a customer of the DMS;

capture, by the DMS, a first snapshot of a computing system associate with the customer;

perform a check, by the DMS and based at least in part on capturing the first snapshot, of a set of files within the first snapshot based at least in part on the set of malware detection rules, wherein the set of files comprise files that have been modified or added with respect to a second snapshot of the computing system prior in time to the first snapshot;

detect, by the DMS and based at least in part on the check, that a file of the set of files comprises first malware; and

identify, by the DMS based at least in part on detecting that the file comprises the first malware, a temporally first snapshot of the computing system that includes the file comprising the first malware.

18. The apparatus of claim **17**, wherein the one or more processors are individually or collectively further operable to execute the code to cause the apparatus to:

transmit, by the DMS to a computing device associated with an administrative account associated with the customer, an alert message comprising an indication that the file comprises the first malware and an indication of the temporally first snapshot.

19. The apparatus of claim **18**, wherein the one or more processors are individually or collectively further operable to execute the code to cause the apparatus to:

receive, by the DMS from the computing device and based at least in part on the alert message, a recovery request message, wherein the recovery request message comprises an indication to perform a recovery procedure for the computing system using a recovery snapshot that is temporally prior to the temporally first snapshot; and

perform, by the DMS, the recovery procedure based at least in part on the recovery request message.

20. A non-transitory computer-readable medium storing code, the code comprising instructions executable by one or more processors to:

receive, by a data management system (DMS), an indication of a set of malware detection rules associated with a customer of the DMS;

capture, by the DMS, a first snapshot of a computing system associate with the customer;

perform a check, by the DMS and based at least in part on capturing the first snapshot, of a set of files within the first snapshot based at least in part on the set of malware detection rules, wherein the set of files comprise files that have been modified or added with respect to a second snapshot of the computing system prior in time to the first snapshot;

detect, by the DMS and based at least in part on the check, that a file of the set of files comprises first malware; and

identify, by the DMS based at least in part on detecting that the file comprises the first malware, a temporally first snapshot of the computing system that includes the file comprising the first malware.

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